

an argument against eclogite being a major component of the upper mantle is that, by contrast with peridotite, it does not contain olivine and would not yield the observed anisotropic P_n velocities. However, the gabbro-to-eclogite transformation may occur in subducting slabs (Section 5.4.2) and play a role in causing earthquakes there.

3.3 Reflection seismology

In the last section, we concentrated on the use of refracted arrivals to infer velocity structure with depth, and noted that reflected arrivals also contain valuable information for this purpose. Studies using the reflected arrivals, known as *reflection seismology*, determine velocities within the crust, and thus are essential in oil and gas exploration. As a result, data acquisition and processing methods have often been developed first by reflection seismologists. For example, digital data were generally used in exploration before they became common in earthquake studies. Similarly, because reflection data are densely sampled in space and time, and the mathematics of wave propagation in a layered medium is simpler than for a spherical earth, techniques are often first developed with reflection data. In this section we survey basic concepts in reflection seismology, some of which we later apply to earthquakes and the spherical earth.

3.3.1 Travel time curves for reflections

We first consider the simplest geometry: a flat layer of uniform-velocity material underlain by a halfspace with a higher velocity (Fig. 3.2-1). Although most applications use P waves, we write the velocity as “ v ” because the results also apply to S waves. For a layer of thickness h_0 with velocity v_0 , we saw in Section 3.2.2 that the travel time as a function of source-to-receiver distance, known as *offset* in reflection seismology, is

$$T(x)^2 = x^2/v_0^2 + 4h_0^2/v_0^2 = x^2/v_0^2 + t_0^2. \quad (1)$$

The travel time curve $T(x)$ is a hyperbola (Fig. 3.3-1) that intercepts the T axis at $t_0 = 2h_0/v_0$, the travel time at zero offset. This time is called the two-way vertical travel time, because the corresponding ray traveled vertically down to the reflector and back. Although this curve is the same as the “reflected wave” curve in Fig. 3.2-2, the convention in reflection seismology is to plot time increasing downward,¹ because later arrivals reflect deeper in the earth.

The layer velocity is found from the slope of the hyperbola. Because the slope decreases with increasing velocity, “flatter” travel time curves indicate higher velocities. To see this, note that a plot of $T(x)^2$ versus x^2 has slope $1/v_0^2$. Alternatively, the variation in travel time with offset is often stated in terms of

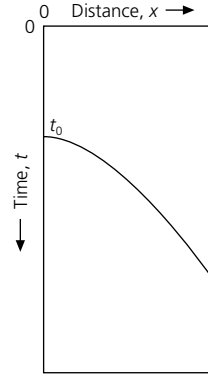


Fig. 3.3-1 The travel time curve for a reflection off a flat interface is a hyperbola, with the minimum at $x = 0$ corresponding to a vertical ray path. The slope is zero at $x = 0$ and increases with the offset distance.

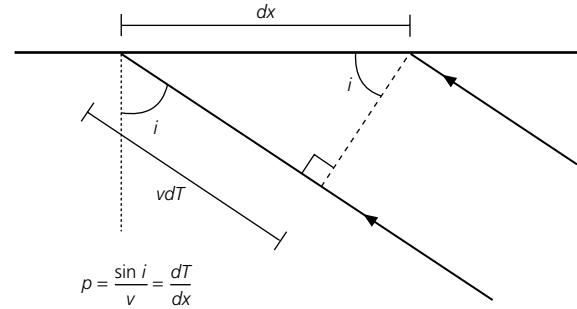


Fig. 3.3-2 Two rays showing the relationship between the angle of incidence, ray parameter, and the slope of the travel time curve for a flat medium.

normal moveout (NMO), the difference between the travel time at some offset and that at zero offset,

$$T(x) - t_0 = (x^2/v_0^2 + t_0^2)^{1/2} - t_0. \quad (2)$$

Once the velocity is found, the layer thickness is given by the vertical travel time.

To see the relation between the travel time curve and ray paths, consider the ray paths to two points dx apart, which differ in travel time by dT (Fig. 3.3-2). Because the ray paths differ in length by vdT , the angle of incidence can be found using

$$\sin i = \frac{vdT}{dx} \quad (3)$$

or, in terms of the ray parameter p (Section 2.5.7),

$$p = \frac{\sin i}{v} = \frac{dT}{dx}. \quad (4)$$

This is consistent with our earlier definition of the ray parameter as the reciprocal of the apparent velocity along the

¹ Earthquake seismologists generally follow the opposite convention.

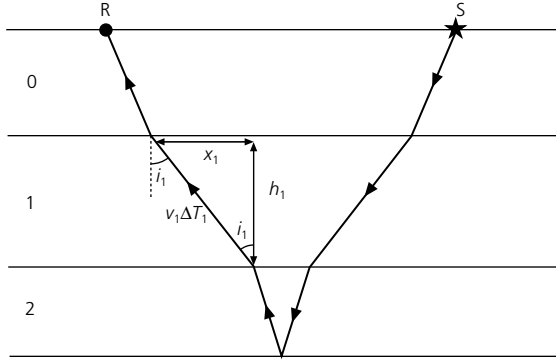


Fig. 3.3-3 Ray geometry for a reflection in a flat-layered medium. Layer thicknesses are h_j , horizontal distances traveled in the layers are x_j , and one-way travel times spent in the layers are ΔT_j .

surface of the wave front, which moves a distance dx in time dT , because

$$p = 1/c_x = 1/(dx/dT). \quad (5)$$

Thus the ray parameter and the angle of incidence of the ray emerging at a distance x can be found from dT/dx , the slope of the travel time curve evaluated at x . From Eqn 2, the slope is zero at $x = 0$ and then increases with offset; so the angle of incidence is nearly zero (vertical incidence) at short distances and becomes closer to 90° (horizontal) at larger distances (Fig. 3.3-1).

This lets us find the travel time curve for reflections in a geometry with multiple horizontal layers. Figure 3.3-3 shows that the reflection R_{n+1} from the top of the $(n+1)^{\text{th}}$ layer (or the bottom of the n^{th} layer) has traveled through n layers, each of thickness h_j and velocity v_j . Such rays, which have been reflected only once, are known as primary reflections. Because, by Snell's law, the ray parameter p is constant along a ray, the incidence angles i_j in each layer can be found from the incidence angle i_0 in the top layer,

$$p = \frac{\sin i_j}{v_j} = \frac{\sin i_0}{v_0}. \quad (6)$$

A downgoing ray, which travels a horizontal distance x_j in the j^{th} layer, spends a time ΔT_j in the layer. Thus, in going down and up again, the ray travels a total horizontal distance

$$x(p) = 2 \sum_{j=0}^n x_j = 2 \sum_{j=0}^n h_j \tan i_j \quad (7)$$

in a total time

$$T(p) = 2 \sum_{j=0}^n \Delta T_j = 2 \sum_{j=0}^n \frac{h_j}{v_j \cos i_j}. \quad (8)$$

We explicitly write $x(p)$ and $T(p)$, because the two sums are formulated in terms of the ray parameter. To see how they

allow us to compute the corresponding travel time curve $T(x)$, consider a single layer, where $x_0 (= x/2)$ is the horizontal distance along each of the downgoing and upgoing legs. In this case, Eqn 8 becomes

$$T(x) = 2[(x/2)^2 + h_0^2]^{1/2}/v_0, \quad (9)$$

because

$$\cos i_0 = h_0(x_0^2 + h_0^2)^{-1/2}. \quad (10)$$

Hence Eqn 8 yields Eqn 9, which is equivalent to the relation we derived earlier showing that the travel time curve for the reflection is a hyperbola (Eqn 1).

For multiple layers, we approximate the travel time curve for the reflection R_{n+1} off the top of the $(n+1)^{\text{th}}$ layer as a hyperbola,

$$T(x)^2_{n+1} = x^2/\bar{V}_n^2 + t_n^2, \quad (11)$$

and find the two parameters, $\bar{V}_n$ and t_n . t_n is the total two-way (up and down) vertical travel time at zero offset, which is twice the sum of the one-way vertical travel times Δt_j for each layer

$$t_n = 2 \sum_{j=0}^n \Delta t_j = 2 \sum_{j=0}^n (h_j/v_j). \quad (12)$$

The velocity term, $\bar{V}_n$, is a little trickier. From the geometry, the distance traveled by the downgoing ray in layer j is

$$x_j = v_j \Delta T_j \sin i_j = (v_j^2/v_0) \Delta T_j \sin(i_0), \quad (13)$$

where the last step used Snell's law (Eqn 6). Hence, by Eqn 7, the total distance, x , can be written

$$x = 2 \sum_{j=0}^n x_j = 2 \frac{\sin i_0}{v_0} \sum_{j=0}^n v_j^2 \Delta T_j. \quad (14)$$

Because the ray parameter is constant along a ray, the slope of the travel time curve is, by Eqn 4,

$$\frac{dT}{dx} = \frac{\sin i_0}{v_0} = x/(2 \sum_{j=0}^n v_j^2 \Delta T_j). \quad (15)$$

For the hyperbolic approximation (Eqn 11), the slope of the travel time curve is

$$\frac{dT}{dx} = \frac{x}{\bar{V}_n^2 T}, \quad (16)$$

so we define

$$\bar{V}_n^2 = (2 \sum_{j=0}^n v_j^2 \Delta T_j)/T. \quad (17)$$

Because this was derived for an arbitrary incidence angle, vertical incidence can be used for simplifications, so in each layer the travel time equals the one-way vertical travel time, $\Delta T_j = \Delta t_j$, and the total travel time is $T = 2 \sum_{j=0}^n \Delta t_j$. Hence

$$\bar{V}_n^2 = \left(\sum_{j=0}^n v_j^2 \Delta t_j \right) / \left(\sum_{j=0}^n \Delta t_j \right). \quad (18)$$

$\bar{V}_n$, the appropriate average velocity for the travel time curve, is the time-weighted *root mean square*, or *rms*, velocity for the first n layers. This hyperbolic approximation and the exact solutions agree well except for large offsets.

These results let us find the layer velocities from the travel time curves. Given a reflection from the top of the n^{th} layer, with vertical two-way travel time t_{n-1} and rms velocity $\bar{V}_{n-1}$, and a reflection from the top of the $(n+1)^{\text{th}}$ layer, with vertical two-way travel time t_n and rms velocity $\bar{V}_n$, the velocity in the n^{th} layer is

$$v_n^2 = \frac{\bar{V}_n^2 t_n - \bar{V}_{n-1}^2 t_{n-1}}{t_n - t_{n-1}}. \quad (19)$$

This relationship is called the *Dix equation*.² The resulting velocity, called an *interval velocity*, is better determined for larger offsets, where the slope of the travel time curve is greater. Because the later reflections have higher velocities, and hence flatter travel time curves (Fig. 3.3-4), larger offsets are required to determine velocities at greater depths.

Travel time calculations are more complicated for dipping layers. Figure 3.3-5 shows the geometry for a reflector of dip θ , whose depth along the perpendicular to the reflector below the origin is h . The travel times can be derived using an imaginary source on the line from the surface source normal to the reflector, at the same distance below the layer, so that travel times from the imaginary source to the receivers are the same as from the true source. Applying the law of cosines to triangle RIS shows that

$$\begin{aligned} T^2 &= [x^2 + 4h^2 - 4hx \cos(\theta + \pi/2)]/v_0^2 \\ &= [x^2 + 4h^2 + 4hx \sin \theta]/v_0^2. \end{aligned} \quad (20)$$

This travel time curve is a hyperbola with minimum at $-2h \sin \theta$, so it is not symmetric about $x = 0$. Reflections from a stack of dipping layers yield travel time curves of approximately this form.

It is sometimes useful to think of the earth as having a continuous distribution of velocity with depth, $v(z)$, rather than a stack of discrete layers, each with uniform velocity. The expressions for the ray path and travel time of a ray with ray parameter p for discrete layers can be generalized. The ray path (Fig. 3.3-6) is given by Snell's law, because the ray parameter,

² Named after its discoverer, pioneering exploration seismologist C. Hewitt Dix (1905–84).

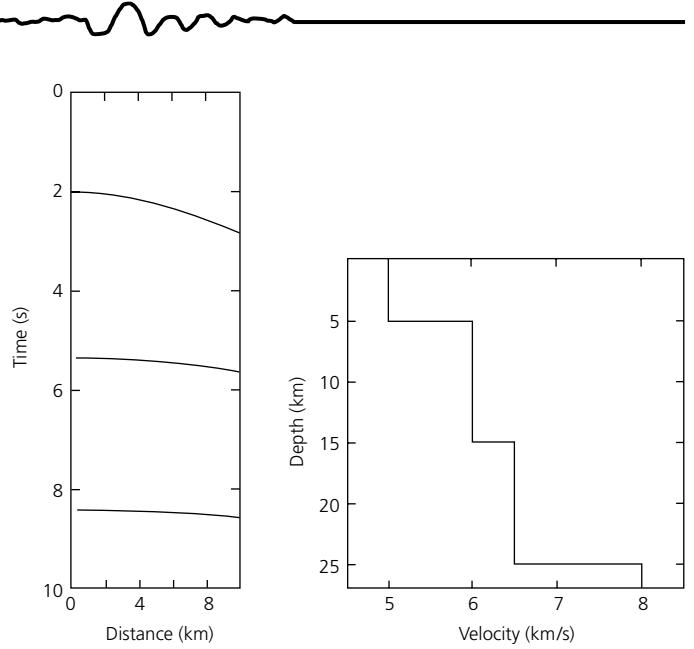


Fig. 3.3-4 Travel time curves for reflections (left) from a layered structure (right) corresponding to continental crust. Reflections from deeper interfaces are flatter, or have shallower slopes, due to the increase of velocity with depth.

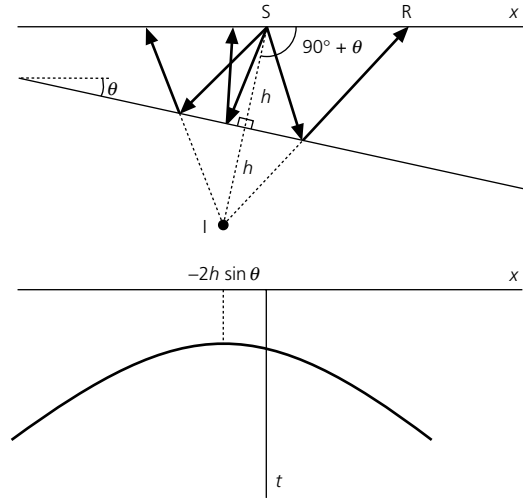


Fig. 3.3-5 The travel time curve for a reflection off a dipping interface can be derived using an imaginary source (I) at depth that gives the same travel times. The resulting hyperbola has a minimum at a nonzero offset. S and R denote the source and the receiver.

$$p = \sin i / v(z), \quad (21)$$

is constant along a ray. If velocity increases with depth, $\sin i$ and thus i increase, so the ray bends away from the vertical on its way down. Once $i = 90^\circ$, the ray turns, becomes horizontal, and then goes upward. At the deepest point, the turning, or bottoming, depth z_p , the velocity is the reciprocal of the ray parameter, $p = 1/v(z_p)$. If on some portion of the ray path the velocity decreases with depth, the ray bends toward the

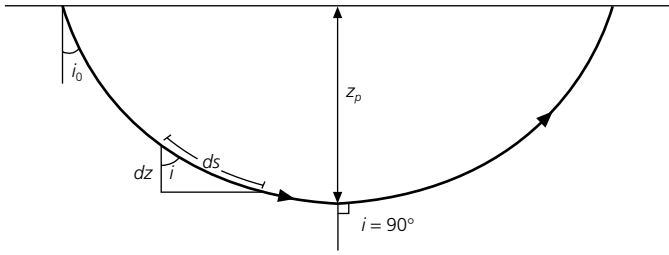


Fig. 3.3-6 Ray path in a medium with velocity increasing smoothly with depth. The ray parameter is constant along the ray path, so the angle of incidence changes as the velocity changes. The incidence angle is smallest at the surface, where velocity is lowest, and is 90° at the bottoming depth, z_p .

vertical. The ray does not turn upward until it gets below the low-velocity region.

We thus replace the sums over layer thickness h_j with integrals over depth, such that the expression for the distance traveled by the ray (Eqn 7) becomes

$$x(p) = 2 \int_0^{z_p} \tan i \, dz = 2p \int_0^{z_p} \left(\frac{1}{v^2(z)} - p^2 \right)^{-1/2} dz, \quad (22)$$

because

$$\sin i = pv(z) \quad \text{and} \quad \cos i = (1 - \sin^2 i)^{1/2} = (1 - p^2 v^2(z))^{1/2}. \quad (23)$$

This is sometimes written in terms of the slowness, the reciprocal of velocity, as

$$u(z) = 1/v(z), \quad (24)$$

so that

$$x(p) = 2p \int_0^{z_p} \frac{dz}{(u^2(z) - p^2)^{1/2}}. \quad (25)$$

Similarly, the travel time sum (Eqn 8) becomes

$$\begin{aligned} T(p) &= 2 \int_0^{z_p} \frac{dz}{v(z) \cos i} = 2 \int_0^{z_p} \frac{dz}{v(z)(1 - p^2 v^2(z))^{1/2}} \\ &= 2 \int_0^{z_p} \frac{u^2(z) dz}{(u^2(z) - p^2)^{1/2}}. \end{aligned} \quad (26)$$

This integral is valid everywhere except at the exact bottom of the curve, where $u(z)$ equals p . A useful way to view this is to note that the ray path (Fig. 3.3-6) can be written as an integral over ds , where $dz = \cos i \, ds$. The travel time is thus

$$T(p) = \int \frac{ds}{v(z)} = \int u(z) ds, \quad (27)$$

the integral of the slowness along the ray path. Slowness, though less intuitive to use than velocity,³ can lead to simpler formulations.

3.3.2 Intercept-slowness formulation for travel times

So far, we have given travel time curves as $T(x)$, the travel time as a function of distance. We now develop an alternative formulation that offers interesting insights and is useful for data analysis. To do so, we note that ΔT_j , the one-way travel time in the j^{th} layer with velocity v_j , is related to the thickness, h_j , and the horizontal distance traveled, x_j (Fig. 3.3-3), by

$$v_j \Delta T_j = (x_j^2 + h_j^2)^{1/2}. \quad (28)$$

The incidence angle i_j for this ray satisfies

$$\sin i_j = \frac{x_j}{(x_j^2 + h_j^2)^{1/2}} = \frac{x_j}{v_j \Delta T_j} \quad \cos i_j = \frac{h_j}{(x_j^2 + h_j^2)^{1/2}} = \frac{h_j}{v_j \Delta T_j}. \quad (29)$$

We rewrite Eqn 28 as

$$v_j \Delta T_j = \frac{x_j^2 + h_j^2}{(x_j^2 + h_j^2)^{1/2}} = x_j \sin i_j + h_j \cos i_j, \quad (30)$$

or

$$\Delta T_j = \frac{x_j \sin i_j}{v_j} + \frac{h_j \cos i_j}{v_j} = p_j x_j + \eta_j h_j, \quad (31)$$

where

$$p_j = (\sin i_j)/v_j = \sin i_j u_j \quad \text{and} \quad \eta_j = (\cos i_j)/v_j = \cos i_j u_j. \quad (32)$$

Thus in layer j we have entities introduced in Section 2.5.7: p_j is the ray parameter, or horizontal slowness, and η_j is the vertical slowness. These are the components of the slowness vector that has magnitude equal to the slowness, and points in the direction of wave propagation. Hence u_j , the slowness in the layer, is

$$u_j^2 = 1/v_j^2 = p_j^2 + \eta_j^2. \quad (33)$$

By Eqn 31, the travel time a ray spends in a layer is the sum of the horizontal slowness times the horizontal distance traveled and the vertical slowness times the vertical thickness. The total travel time is the sum over all layers, with a factor of two to account for both downgoing and upgoing legs,

³ It is somehow harder to think of a zone of high slowness than a low-velocity zone.

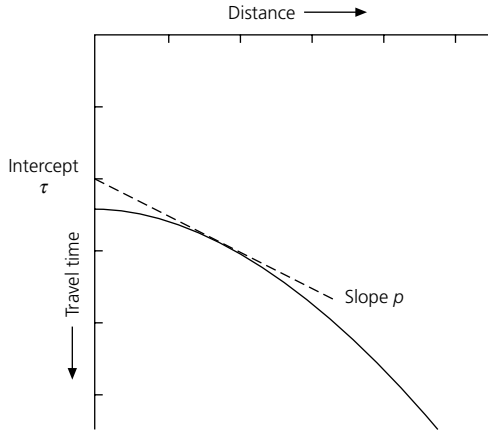


Fig. 3.3-7 Relation between the travel time curve $T(x)$ and the line tangential to a point on it, which has a slope, or slowness, p and a time axis intercept τ .

$$T(x) = 2 \sum_{j=0}^n \Delta T_j = 2 \sum_{j=0}^n p_j x_j + 2 \sum_{j=0}^n \eta_j h_j. \quad (34)$$

By Snell's law, the horizontal ray parameter is constant along the ray path, so $p_j = p$, and

$$T(x) = px + 2 \sum_{j=0}^n \eta_j h_j, \quad (35)$$

where $x = 2 \sum_{j=0}^n x_j$ is the total horizontal distance traveled. This formulation is equivalent to the way we formulated the travel time as the scalar product of the distance and slowness vectors (Eqn 2.5.34).

Formulating the travel time curve in this way gives interesting insight. We define

$$T(x) = px + \tau(p), \quad (36)$$

where the function

$$\tau(p) = 2 \sum_{j=0}^n \eta_j h_j = 2 \sum_{j=0}^n (1/v_j^2 - p^2)^{1/2} h_j = 2 \sum_{j=0}^n (u_j^2 - p^2)^{1/2} h_j. \quad (37)$$

Because p is the slope of the travel time curve (dT/dx) and hence of a line tangential to it at the point (T, x) , τ is the intercept of the tangent line with the time axis (Fig. 3.3-7). In general τ and p differ for different points on the travel time curve, so the travel time curve can be described by the values of either (T, x) or (τ, p) . Thus the function $\tau(p)$ is called the *intercept-slowness* representation of the travel time curve. Although less intuitive, the $\tau(p)$ formulation is equivalent to $T(x)$.

Given that the slope of the travel time curve $T(x)$ has special significance, it is natural to investigate the slope of the function $\tau(p)$. To do this, we write Eqn 36 with the ray parameter, rather than the distance, as the independent variable,

$$\tau(p) = T(p) - px(p), \quad (38)$$

and differentiate

$$\frac{d\tau}{dp} = \frac{dT}{dp} - p \frac{dx}{dp} - x(p) = \frac{dT}{dx} \frac{dx}{dp} - p \frac{dx}{dp} - x(p) = -x(p). \quad (39)$$

Thus, just as p is the slope of the travel time curve, $T(x)$, the distance, x , is minus the slope of the $\tau(p)$ curve.

To illustrate these ideas, we show that the $\tau(p)$ formulation gives the travel time curve for the reflected wave in a layer over a halfspace. Figure 3.3-3 shows that $x_0 = x/2$, so, using Eqn 32,

$$p = \frac{(x/2)}{v_0[(x/2)^2 + h_0^2]^{1/2}}, \quad \eta_0 = \frac{h_0}{v_0[(x/2)^2 + h_0^2]^{1/2}}. \quad (40)$$

Hence, by Eqns 36 and 37, the travel time curve is

$$\begin{aligned} T(x) &= px + 2\eta_0 h_0 = \frac{(x/2) + 2h_0^2}{v_0[(x/2)^2 + h_0^2]^{1/2}} \\ &= 2[(x/2)^2 + h_0^2]^{1/2}/v_0, \end{aligned} \quad (41)$$

which is the familiar hyperbola (Eqn 9).

To see how this travel time curve appears when written as $\tau(p)$, we write Eqn 37 for a layer over a halfspace:

$$\tau(p) = 2(1/v_0^2 - p^2)^{1/2} h_0. \quad (42)$$

This can also be written as

$$(v_0^2 \tau^2)/(4h_0^2) + v_0^2 p^2 = 1, \quad (43)$$

which is an ellipse whose axes are the τ and p axes (Fig. 3.3-8). It intersects the τ axis at $(\tau = t_0 = 2h_0/v_0, p = 0)$, and the p axis at $(\tau = 0, p = 1/v_0)$. Both these points have significance. The first, where the travel time curve has zero slope and the time axis intercept is the vertical two-way travel time, corresponds to the zero-offset point $x = 0$.

The second, where the travel time curve has slope $1/v_0$ and time axis intercept 0, is the $\tau(p)$ position of the linear travel time curve for the direct wave. Hence the line for the direct wave maps to a point in the $\tau(p)$ plane that is on the ellipse describing the reflected wave. To understand why this occurs, we use the fact that distance is minus the derivative of the $\tau(p)$ curve (Eqn 39) and differentiate Eqn 42, giving

$$x(p) = -d\tau/dp = 2ph_0(1/v_0^2 - p^2)^{-1/2}, \quad (44)$$

so at the point $p = 1/v_0$, $x = \infty$. This makes sense, because as $x \rightarrow \infty$, the reflected wave is asymptotic to the direct wave (Fig. 3.2-2).

The head wave is easily mapped into the $\tau(p)$ plane, because its travel time curve (Eqn 3.2.8) is

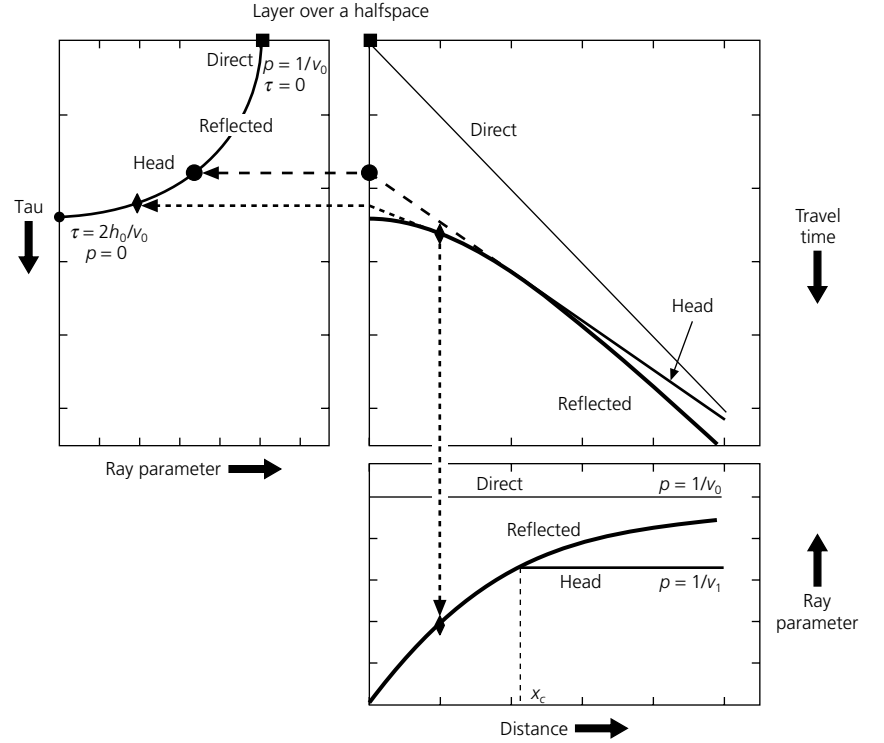


Fig. 3.3-8 Travel time curves $T(x)$ for a layer over a halfspace and their representation in the (τ, p) plane. Each point on the $T(x)$ curves has a slope (ray parameter) p and intercept τ . The linear travel time curves for the direct and head waves each map into a point (square and circle) in the (τ, p) plane. The hyperbolic travel time curve for the reflection maps into an ellipse in the (τ, p) plane. Note how an arbitrary point on the reflection's travel time curve, marked by the diamond, maps into the other two curves.

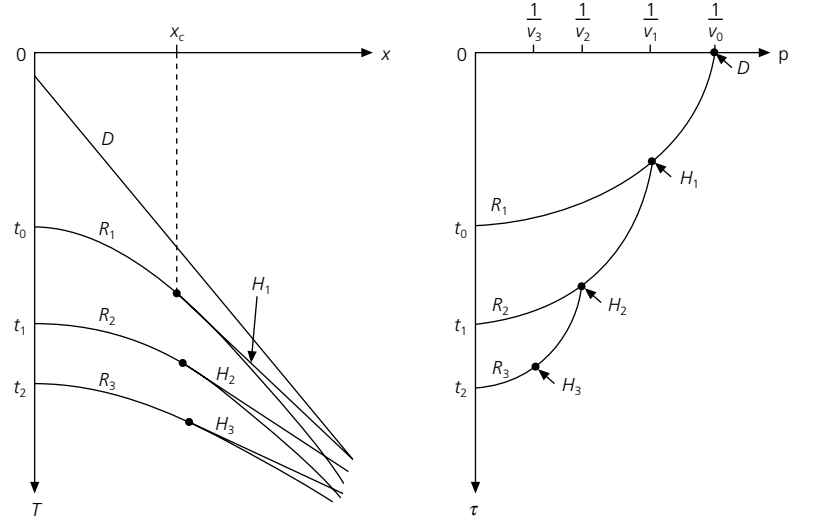


Fig. 3.3-9 Relation between the travel time curve $T(x)$ and the function $\tau(p)$ for multiple layers over a halfspace. D denotes the direct wave; R_i and H_i are reflections and head waves at the top of the i^{th} layer; x_c is the critical distance for H_1 . (After Diebold and Stoffa, 1981. Reproduced by permission of the Society of Exploration Geophysicists.)

$$T_H(x) = x/v_1 + 2h_0(1/v_0^2 - 1/v_1^2)^{1/2} \\ = x/v_1 + \tau_1, \quad (45)$$

a line with slope equal to the reciprocal of the halfspace velocity, $p = 1/v_1$, and intercept τ_1 . Thus the head wave maps into a point on the ellipse describing the reflected wave, corresponding to the critical distance x_c where the head and reflected waves are the same. To see this, note that for $p = 1/v_1$, Eqn 44 gives

$$x(p) = -d\tau/dp = 2h_0v_0(v_1^2 - v_0^2)^{-1/2} = x_c. \quad (46)$$

This point divides the ellipse describing the reflected wave into a subcritical portion, between the τ axis and the head wave, and a postcritical portion, between the head wave and the p axis. We will see shortly that the fact that different arrivals have distinct locations in the $\tau(p)$ plane provides the basis for techniques that can separate these arrivals.

This analysis can be extended to more complex geometries. For multiple layers, the $\tau(p)$ curves corresponding to reflections off successive layers are all portions of different ellipses (Fig. 3.3-9). For a continuous velocity distribution, the summation for τ (Eqn 37) becomes an integral

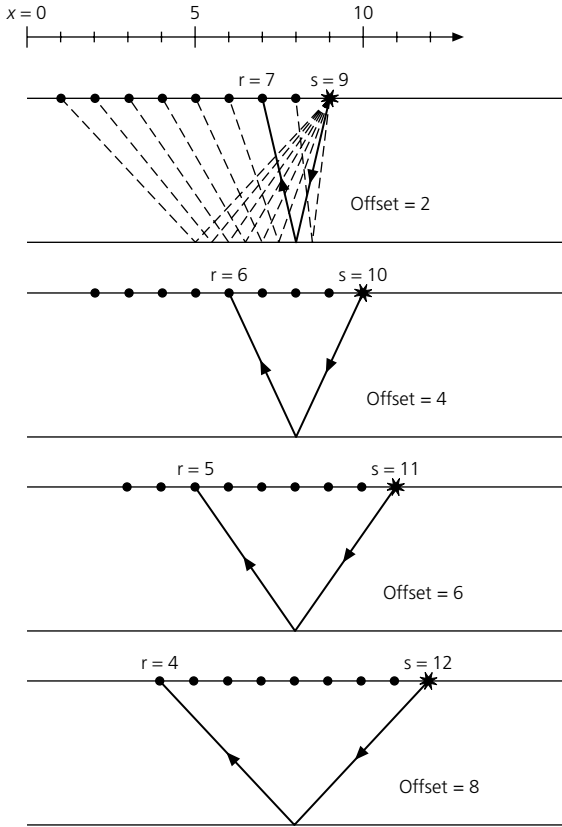


Fig. 3.3-10 Schematic geometry of a multichannel seismic reflection survey with a single source (star) and eight receivers (dots) moving along a survey line. Each physical experiment produces eight seismograms corresponding to ray paths (dashed lines) with a single source location and a range of receiver locations. Four seismograms from different source and receiver positions, corresponding to the ray paths shown by solid lines, sample the same point at depth on a flat reflector. These have the same midpoint halfway between source and receiver, but different source-to-receiver offsets.

$$\tau(p) = 2 \int_0^{z_p} \eta(z) dz = 2 \int_0^{z_p} (1/v^2(z) - p^2)^{1/2} dz = 2 \int_0^{z_p} (u^2(z) - p^2)^{1/2} dz. \quad (47)$$

Formulating travel time curves as $\tau(p)$ is useful for some techniques that invert for velocity structure.

3.3.3 Multichannel data geometry

A feature of reflection seismology is *multichannel* geometry, the use of multiple source and receiver locations, so that points on reflecting interfaces are sampled repeatedly. Figure 3.3-10 illustrates how such coverage is accomplished by combining experiments performed with a seismic source and an array of eight receivers at fixed distances from the source. Each time the source is activated, eight seismograms, or *traces*, are recorded.

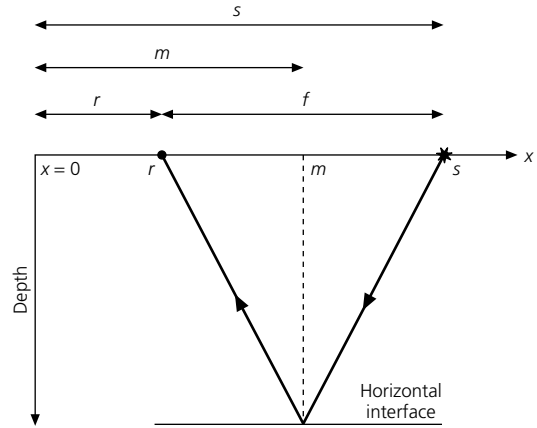


Fig. 3.3-11 Relation between source, receiver, midpoint, and offset coordinates measured along the survey line. Any two specify an individual seismogram.

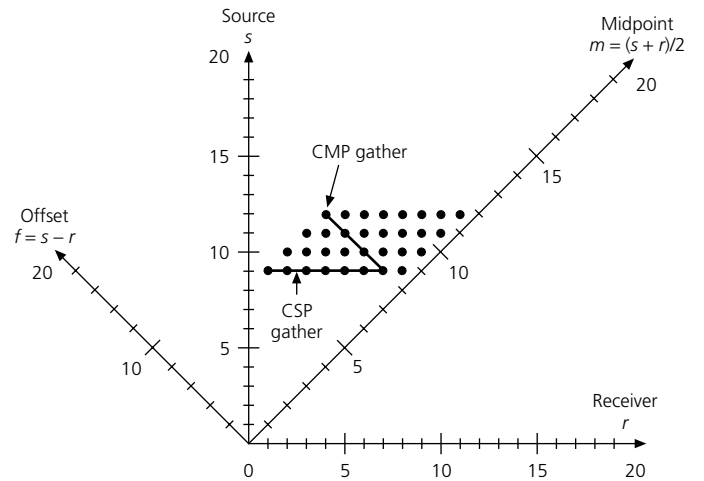


Fig. 3.3-12 An individual trace is characterized by its position in a two-dimensional diagram showing its source, receiver, midpoint, and offset coordinates. Dots show the traces indicated in Fig. 3.3-10. Physical experiments correspond to a common source point (CSP) gather; the four traces in Fig. 3.3-10 with the same midpoint form the common midpoint (CMP) gather shown.

The source and the receivers are then moved, and the experiment is repeated, giving eight more traces. Eventually each point on the reflector is sampled four times, producing “four-fold coverage.”

We assume initially that the velocity structure is layered and varies only with depth. Even so, the four seismograms that sample the same point are not identical, because they correspond to different source and receiver positions, and thus different offset distances between the source and the receiver. Hence each trace is a record of displacement, or pressure, as a function of time, t , $u(s, r, t)$, characterized by the source and receiver positions.

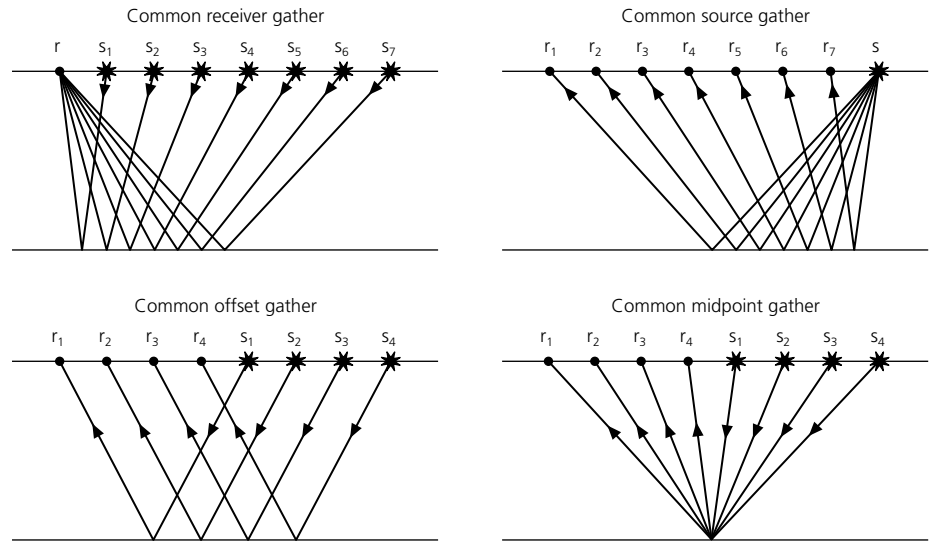


Fig. 3.3-13 Schematic of the four different gather types.

The data are analyzed by grouping the seismograms that sampled the same point on the reflector. In this flat-layered geometry, these seismograms have the same point, known as the *midpoint*, halfway between the source and the receiver. For each midpoint, there is a set of traces with different offsets. The midpoint m and offset f are defined in terms of the source location s and the receiver position r as

$$m = (s + r)/2, \quad f = (s - r). \quad (48)$$

Thus an individual seismogram is specified by either the source and receiver positions, or the midpoint and offset (Fig. 3.3-11). These are plotted using two perpendicular axes (Fig. 3.3-12), one for the source location and one for the receiver position. The midpoint and offset for each seismogram are indicated by distance along axes 45° from the s and r axes. Note that the scales on these axes differ from the other two.

To illustrate this relationship, consider the four experiments in Fig. 3.3-10, with eight receivers and a single source. Each experiment produced data at points, shown by dots, with constant source position and successive receiver positions. Successive experiments yielded data along a similar horizontal line, but displaced by the motion of the source and the receiver.

The data can be sorted and combined in various ways that need not correspond to an actual experiment (Fig. 3.3-13). Each experiment corresponds to a set of records with the same source position, a *common source point*, or CSP, gather. Traces with the same midpoint and different offsets can be grouped in a *common midpoint*, or CMP, gather. Similarly, common receiver point and common offset gathers can be formed.

Ordering traces by midpoint and offset makes no distinction between a source at position a and a receiver at position b , or the reverse. This assumption is justified by the principle of reciprocity, by which these two geometries should produce identical seismograms. Thus a common receiver point gather

can simulate a reversed profile (Section 3.2.2) because, by reciprocity, it gives the same data as a common source point gather shot in the opposite direction.

Later in this section, we will discuss a few aspects of the data collection process. The sources can be explosives, sound sources in water, or vibration sources on land. The source coordinate is thus sometimes referred to as a source point, shot point, or vibration point. The receivers are typically single-component vertical seismometers, known as geophones, for land applications, and pressure transducers, or hydrophones, for marine surveys. The receiver coordinate is thus often termed the geophone coordinate. Generally large numbers of receivers, which are themselves groups of receivers, are used. Increasingly, data are collected over two-dimensional areas, and so are processed to yield three-dimensional velocity structures.

3.3.4 Common midpoint stacking

Because the traces in a CMP gather have ideally sampled the same subsurface point with different offsets, they can be combined to enhance reflected arrivals. The process begins with a set of traces showing the data as a function of offset and time. The data contain “signals” of interest, primary reflections from interfaces that are used to determine velocity structure with depth. The data also contain “noise,” arrivals of no interest, including direct waves, head waves,⁴ surface waves (sometimes termed “ground roll”), and waves from the source that travel in the air. The data may also contain arrivals (Fig. 3.3-14) that have been reflected more than once, which are known as *multiples*, by contrast with the once-reflected primary reflections.

To enhance primary reflections and suppress everything else, we exploit the fact that the arrival times of various signals

⁴ In the previous section we focused on direct and head waves, illustrating the adage that “one person’s signal is another’s noise.”

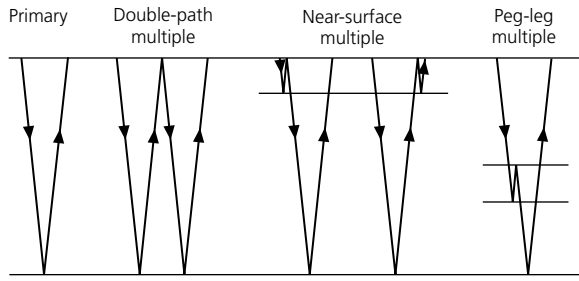


Fig. 3.3-14 Geometry of various multiple reflections. (After Kearey and Brooks, 1984.)

vary in different ways between traces as a function of offset (Fig. 3.3-15). Reflections have hyperbolic travel time curves, whereas direct waves, head waves, surface waves, and air waves have linear travel time curves. Other noise may be essentially incoherent between traces.

Consider a reflection whose variation in travel time with offset is the normal moveout (NMO),

$$T(x) - t_0 = (x^2/\bar{V}^2 + t_0^2)^{1/2} - t_0, \quad (49)$$

where t_0 and $\bar{V}$ are the vertical two-way time and rms velocity. If each trace is shifted forward in time by the appropriate NMO, this reflection appears at the same time for all offsets (Fig. 3.3-15). By contrast, arrivals with different moveouts, such as the direct wave, do not align. Similarly, multiple reflections do not align, because they reflected off shallower interfaces than primary reflections with a similar arrival time, and thus have a lower rms velocity. This method is similar to forming reduced travel time plots (Section 3.2), where a linear time shift lines up direct or head waves whose linear travel time curve has apparent velocity equal to the reducing velocity. In this case, the hyperbolic time shift lines up reflections with hyperbolic travel time curves.

If the traces are added after this time shift, the resulting sum, in theory, is the single trace that would have been recorded at zero offset, with coincident source and receiver. The reflection

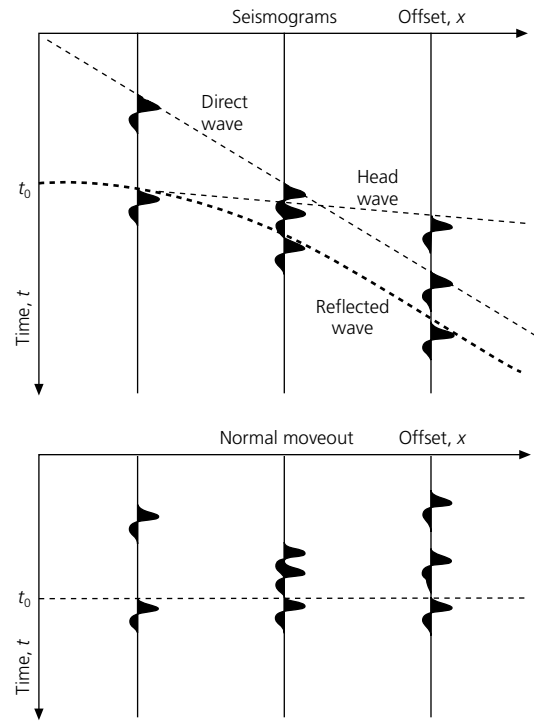


Fig. 3.3-15 Schematic example of the normal moveout correction, shown for the three arrivals for a single layer. NMO aligns all traces (lower panel) in a common midpoint gather by a time shift corresponding to the hyperbolic travel time curve of a reflection. The desired reflection is thus in phase between traces, whereas other arrivals are out of phase. CMP stacking, which adds the traces after this time shift, enhances the desired reflection and suppresses other arrivals.

that was aligned is in phase on all traces, and thus sums constructively and gives a strong arrival. By contrast, other arrivals will have been shifted such that they are sometimes out of phase, and thus sum destructively, yielding weaker arrivals. The process of time shifting and then summing the traces with different offsets for a given midpoint is called *common midpoint (CMP) stacking*.

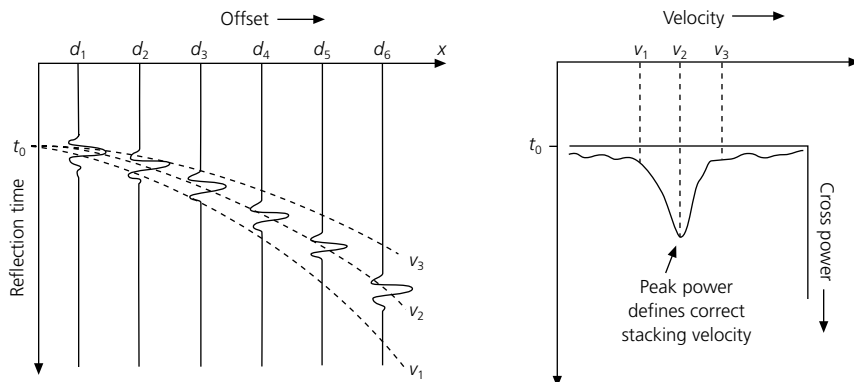


Fig. 3.3-16 Schematic of CMP stacking and velocity analysis. *Left*: Stacking is done for a range of stacking velocities, each corresponding to a different hyperbola in offset–time space. *Right*: The peak in the velocity spectrum, or power in the resulting stack, shows the best stacking velocity. (After Taner and Kohler, 1969. Reproduced by permission of the Society of Exploration Geophysicists.)

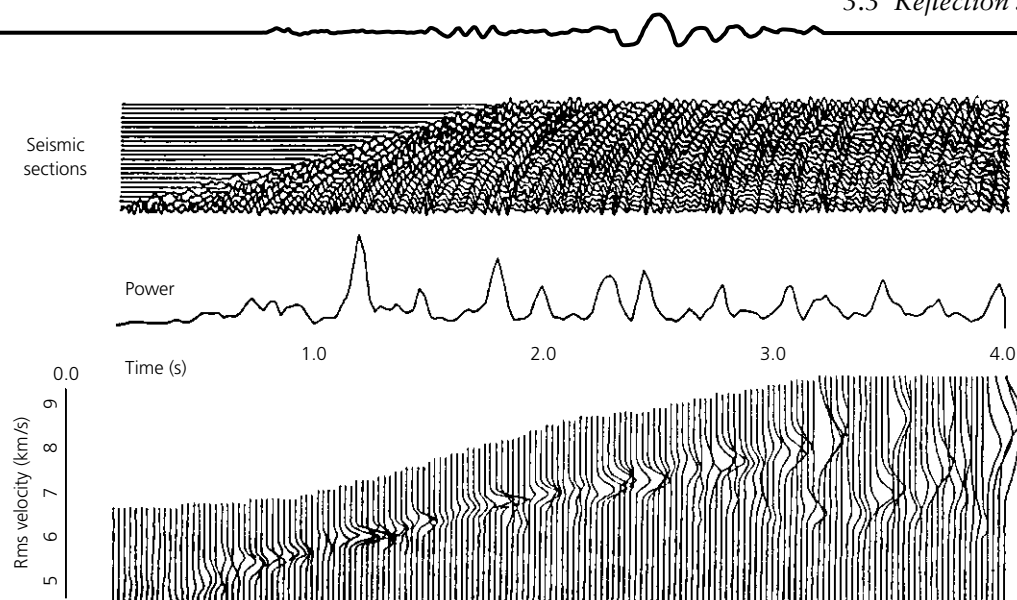


Fig. 3.3-17 Example of CMP stacking and velocity analysis. Velocity analysis at different times yields the best stacking velocity as a function of time (*bottom*). The stacking velocity increases with time because later arrivals reflected off deeper interfaces. (After Taner and Kohler, 1969. Reproduced by permission of the Society of Exploration Geophysicists.)

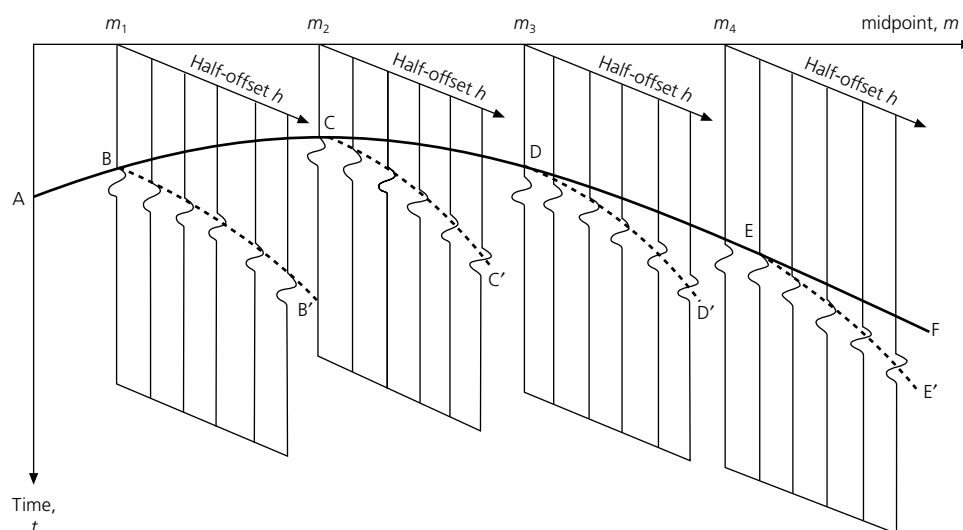


Fig. 3.3-18 Schematic geometry illustrating formation of a zero-offset section by common midpoint stacking. Each CMP gather is stacked over all offsets, as shown by the dashed lines like B-B', to produce a single zero-offset trace for that midpoint. Taken together, these traces form a zero-offset section, a plane in midpoint-time space, containing arrivals like that shown by the solid curve A-F. (After Robinson, 1983. *Migration of Geophysical Data*, © 1983, p. 24. Reprinted by permission of Pearson Education.)

Real data contain more than one reflection, and the appropriate velocities are unknown. Thus the velocities are found by stacking with a range of velocities and determining which gives the best results. As illustrated in Fig. 3.3-16, traces are stacked along hyperbolas corresponding to different velocities. The stack output as a function of stacking velocity, known as a velocity spectrum, has peak amplitude at the velocity that best aligns arrivals on the different traces. This stacking velocity is close to the rms velocity if the data are reasonably good and the structure is approximately a set of flat layers.

Because later reflections have higher rms velocities, they yield higher stacking velocities. Thus velocity analysis is conducted as a function of time. In Fig. 3.3-17, the best stacking

velocity, indicated by the maximum in the velocity spectrum, increases with time for deeper arrivals. This increase “tunes” the stacking to bring arrivals with various stacking velocities “into focus.” Peaks in the power of the velocity spectrum show the arrival of strong coherent reflections. At later times there are several peaks, as multiples arrive. Using the stacking velocities, interval velocities for different depths are found from the Dix equation.

Figure 3.3-18 illustrates the CMP concept geometrically. The traces give displacement or pressure as a function of midpoint, offset, and time, $u(m, f, t)$. CMP gathers can be thought of as planes parallel to the offset and time axes, each with the appropriate midpoint. Each gather is stacked over all offsets

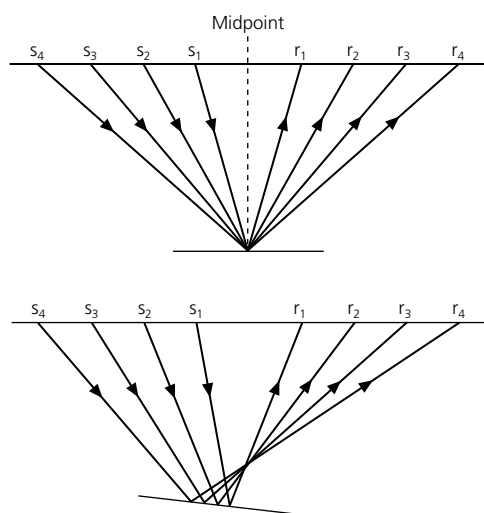


Fig. 3.3-19 *Top:* Traces with a common midpoint sample the same point on a reflector when a reflector and the structure above it are horizontal. *Bottom:* If the structure dips, traces with a common midpoint do not reflect at the same point. (After Kearey and Brooks, 1984.)

to produce a zero-offset trace for that midpoint. These traces together form a zero-offset *seismic section*, $u(m, 0, t)$, a function of midpoint and time. This section simulates moving along the survey line with a single source and receiver at the same location, and recording arrivals from below as a function of time. Because this process reduces the volume of data dramatically, there is a tendency to conduct processing operations after, rather than before, stacking when possible.

Often a CMP stack is referred to as a CDP, or common depth point, stack. CMP is a better term, because traces with the same midpoint have the same reflection point at depth only when a reflector and the structure above it are flat-lying (Fig. 3.3-19). This effect is generally small enough that CMP stacking is useful. We will discuss shortly the limitations on reflection studies due to deviations from the ideal flat geometry.

A seismic section is in some ways similar to a “picture” of the subsurface. Major arrivals in the data generally represent significant reflectors at depth, and can be correlated with geologic structure. As a result, analysis of seismic reflection data is a powerful geological tool. For example, Fig. 3.3-20 (*top*) shows a seismic section across the Peru trench. Data of one polarity are black, making coherent reflectors more visible. The interpretation (*bottom*) indicates the top of the crust of the subducting Nazca plate, including small grabens, and complex structures in the overlying accretionary prism.

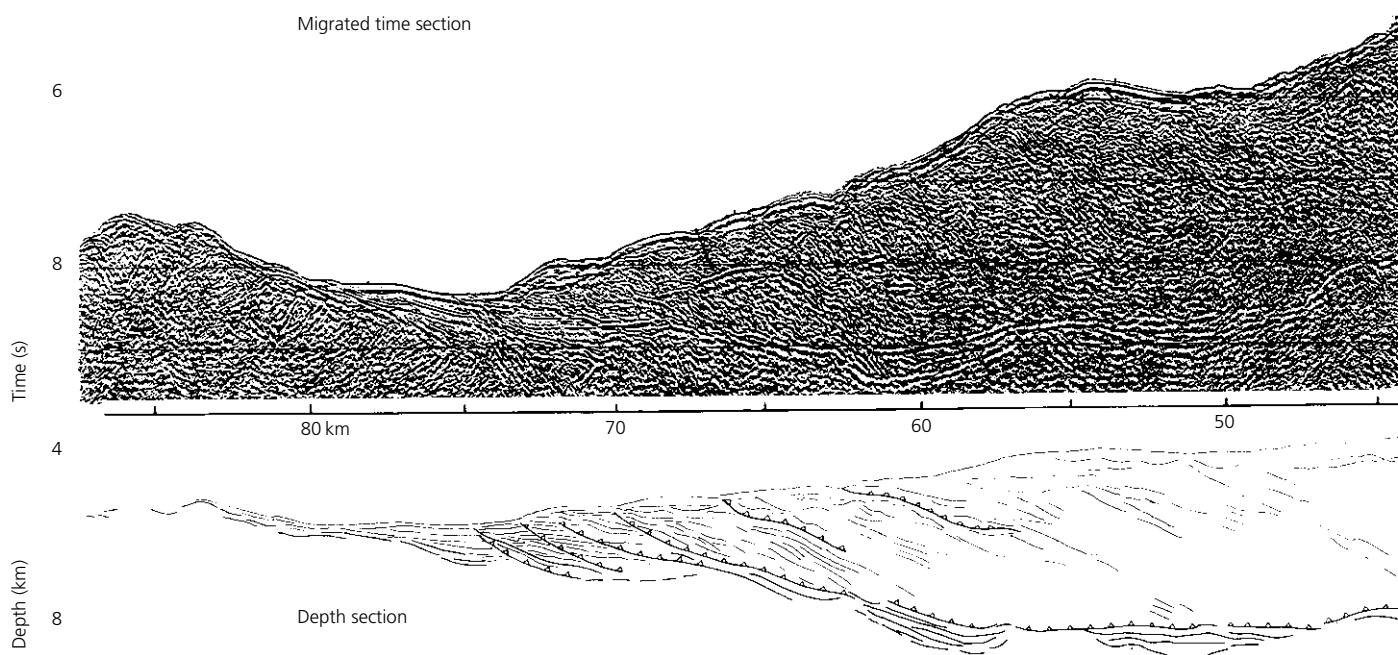


Fig. 3.3-20 Migrated seismic section across the Peru trench, showing the subducting Nazca plate dipping to the right. The data were collected with air gun sources shot at 35 m intervals and recorded by a 1600 m-long array with 24 hydrophone groups. The data were sampled every 4 milliseconds. (After Von Huene *et al.*, 1985. *J. Geophys. Res.*, 90, 5429–42, copyright by the American Geophysical Union.)

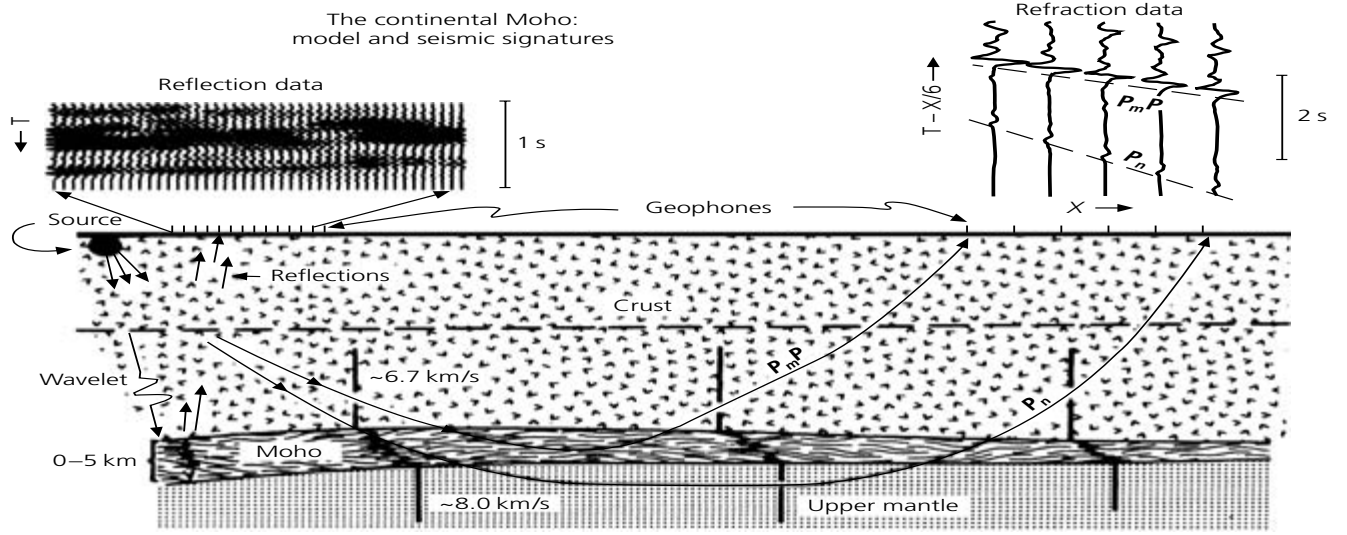


Fig. 3.3-21 Reflection data (left), showing muting (right) to eliminate the head waves that arrive first and the large surface waves that arrive later. (After Claerbout, 1985.)

3.3.5 Signal enhancement

The best hope of reducing artifacts in a seismic section due to noise and other difficulties is to exclude them before stacking. Thus, as in many signal processing applications, the idea is to identify characteristics of the “noise” we seek to reject, and use those characteristics to exclude it.

For example, variations in the thickness of a near-surface low-velocity layer due to weathering produce arrival time variations. Similar variations can result from sea floor topography, because the water is a low-velocity material of varying thickness, or from elevation changes along a land survey. These shifts can cause the travel time of reflections to deviate from the hyperbolic moveout with offset assumed in stacking, and hence degrade a stacked section and produce spurious relief on a deeper reflector. To minimize these problems, a *static time correction*, shifting traces back or forward in time, can be applied.

Direct waves, head waves, surface waves, air waves, and the like are often identifiable on CSP gathers from their arrival times and linear travel time curves. Data corresponding to the time–distance ranges in which the undesired arrivals appear can be set equal to zero, or *muted* before the gathers are stacked (Fig. 3.3-21).

Another approach to isolating reflections uses the fact that the apparent velocity along the surface,

$$c_x = 1/p = v/\sin i = \omega/k_x, \quad (50)$$

is higher for reflections, which have angles of incidence close to the vertical, than for surface or air waves. Hence the reflections have a longer apparent wavelength along the surface, $\lambda_x = 2\pi c_x/\omega$. Thus the effects of surface waves can be reduced by summing a *group* of receivers to produce a single trace. Arrivals

with wavelengths shorter than the length of the group interfere destructively and are reduced in amplitude, enhancing the longer-wavelength reflections. Hence traces from a single source–receiver pair are often actually a sum of a number of geophones or hydrophones. In this way, the data collection process, rather than subsequent analysis, enhances the reflections.

Differences in the apparent velocity can also be used to enhance reflections after the data are collected. In this approach, arrivals with different apparent velocities on common source gathers are separated by *velocity filtering*, using a double Fourier transform. As we saw in Section 2.8.2, and discuss further in Chapter 6, the Fourier transform and inverse transform relate a function of time $f(t)$ and its transform $F(\omega)$, a function of angular frequency,

$$F(\omega) = \int_{-\infty}^{\infty} f(t)e^{-i\omega t} dt \quad f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega)e^{i\omega t} d\omega. \quad (51)$$

Similarly, because the wavenumber is the spatial frequency (Section 2.2.2), it is related to the distance in the same way that angular frequency is related to time. Hence, a function of the horizontal distance $g(x)$ and its corresponding function of horizontal wavenumber $G(k_x)$ are related by the Fourier transform pair

$$G(k_x) = \int_{-\infty}^{\infty} g(x)e^{ik_x x} dx \quad g(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} G(k_x)e^{-ik_x x} dk_x. \quad (52)$$

By convention, opposite signs are used in the exponentials for the time and space transforms.

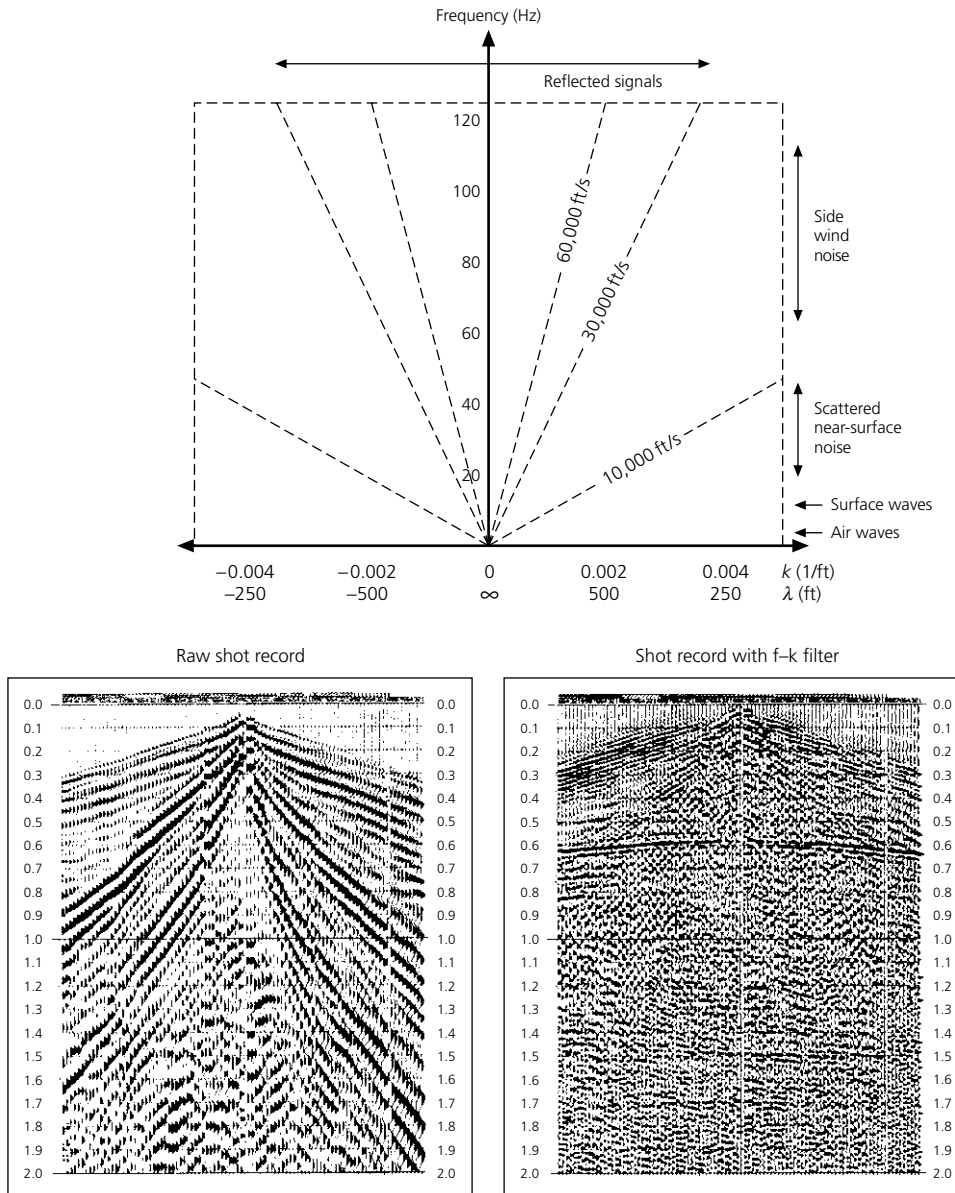


Fig. 3.3-22 Velocity filtering by Fourier transformation into the horizontal wavenumber and frequency domain. *Top*: Positions of reflected waves, noise, air waves, and surface waves in the (k_x, f) plane. Slopes correspond to lines of equal apparent velocity (in ft/s). (After Kanasewich, 1981.) *Bottom*: Common source gather before and after velocity filtering. Surface waves have been suppressed by removing low apparent velocity data, thus enhancing reflections. (Hosking Geophysical.)

A gather $u(x, t)$ is the displacement as a function of horizontal distance and time, so the double Fourier transform,

$$U(k_x, \omega) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} u(x, t) \exp[i(-\omega t + k_x x)] dx dt, \quad (53)$$

converts it to the horizontal wavenumber and angular frequency domains. Plotting the transform as a function of k_x and ω (or, equivalently, k_x and frequency f) separates the data into portions of different apparent velocity, because a given velocity, $c_x = \omega/k_x$, plots as a straight line (Fig. 3.3-22). It is thus possible to suppress arrivals with a given range of apparent velocities by setting the data in some region of (k_x, ω) space

to zero, and inverse transforming the data back to (x, t) space, using the inverse of the double Fourier transform

$$u(x, t) = \frac{1}{4\pi^2} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} U(k_x, \omega) \exp[i(\omega t - k_x x)] dk_x d\omega. \quad (54)$$

Rather than having an abrupt boundary, the data at the edges of the portion of the (k_x, ω) space of interest are tapered smoothly to zero for reasons discussed in Chapter 6.

Thus the double Fourier transform converts data containing arrivals that overlap in the (x, t) domain into the (k_x, ω) domain, where the arrivals have distinct properties that make it easy to separate them. This separation is exploited to filter

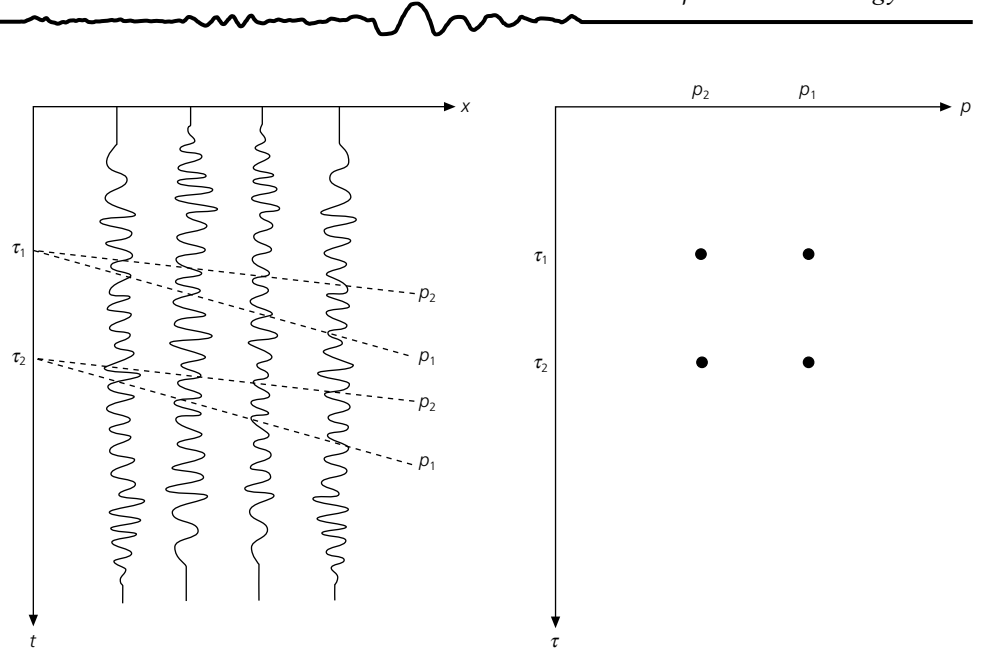


Fig. 3.3-23 Schematic illustration of slant stacking: data are summed along lines in the (x, t) plane (left) corresponding to values of intercept τ and slope p , and so yield points in the (τ, p) plane (right).

the data, which are then transformed back to (x, t) . Velocity filters are also called *dip filters*, because they separate arrivals based on their slope (dip) in the (k_x, ω) domain. A variant of this method for data recorded in two spatial dimensions, $u(x, y, t)$, is to take the triple Fourier transform $U(k_x, k_y, \omega)$. Because the transform is in terms of both components of the horizontal wave vector, it can be filtered to suppress arrivals coming from certain directions.

Another approach to transforming data such that components are more easily separated uses the intercept-slowness formulation of travel time curves. As discussed in Section 3.3.2, the function $\tau(p)$ describes each point on a travel time curve $T(x)$ by the time axis intercept τ and the slope p of the line tangential to the curve at that point. Thus seismic data can be described as functions either of position and time, $u(x, t)$, or of slope and intercept, $\bar{u}(\tau, p)$. To transform from one representation to the other, the data $u(x, t)$ are summed along lines of constant slope in the (x, t) plane, which correspond to values of intercept τ and slope p (Fig. 3.3-23),

$$\bar{u}(\tau, p) = \int_{-\infty}^{\infty} u(x, \tau + px) dx. \quad (55)$$

This integral, which maps all the data along each slanted line in (x, t) to a point in (τ, p) , is called a *slant stack*, or *Radon transform* of the data. It is also called a *plane wave decomposition*, because it decomposes the data according to p , the reciprocal of the apparent velocity of a plane wave. The inverse slant stack operation that transforms the slant stack back into the (x, t) space can be written⁵

$$u(x, t) = 1/t^2 * \frac{1}{2\pi} \int_{-\infty}^{\infty} \bar{u}(t - px, p) dp, \quad (56)$$

where “*” is the convolution operation, discussed shortly. This expression is similar to a slant stack in the (τ, p) plane, because data are summed along a line of constant τ .

All the data are mapped from one domain into the other, so no data are lost by this transformation. Thus, after slant stacking, we can use the fact that the $\tau(p)$ representation of the travel time curve is in some ways simpler than the $T(x)$ representation. Because different arrivals fall in different parts of the (τ, p) plane (Fig. 3.3-8), undesired arrivals can be suppressed by zeroing portions of the data. For example, the gather in Fig. 3.3-24 shows a strong surface wave, the late-arriving linear arrival with an apparent velocity of about 1.35 km/s and intercept about 0. In the usual (x, t) space, it would be hard to filter out this arrival without suppressing the reflections. After slant stacking, this arrival shows up as a region of large amplitude with $\tau \approx 0$ and $p = 1/1350 \text{ s/m} \approx 740 \mu\text{s/m}$. Once the slant stack is filtered by eliminating all data with $p > 650 \mu\text{s/m}$ and inverse transformed, the surface wave is significantly reduced. In practice, rather than having an abrupt boundary, the data at the edges of the portion of the (τ, p) space of interest are tapered smoothly to zero for reasons discussed in Chapter 6.

The slant stack and velocity filtering with the double Fourier transform are related, because both exploit properties of the data associated with the apparent velocity. As a result, slant stacking can be done by transforming data to the (k_x, ω) domain, evaluating the transform for constant values of the ray parameter, and then inverse transforming to the time domain.

⁵ Claerbout (1985).

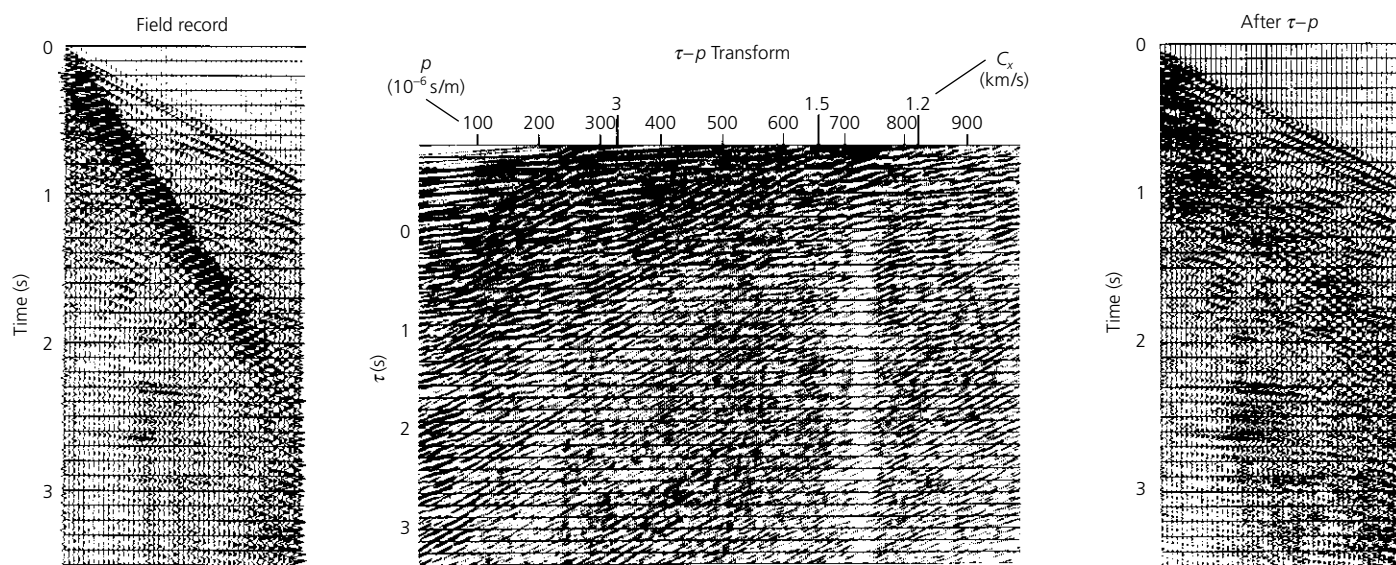


Fig. 3.3-24 *Left*: Common source point gather of Vibroseis data from Alaska, showing prominent late-arriving surface waves with an apparent velocity of about 1.35 km/s and intercept about 0. *Center*: Slant stack of the data. The p axis is labeled both with values of p ($\mu\text{s/m}$) and apparent velocity (km/s). The surface waves appear as a region of large amplitude with $\tau \approx 0$ and $p \approx 740 \mu\text{s/m}$. *Right*: The inverse slant stack, after suppression of data with $p > 650 \mu\text{s/m}$, shows the surface wave significantly reduced. (Tatham, 1989. With kind permission from Kluwer Academic Publishers.)

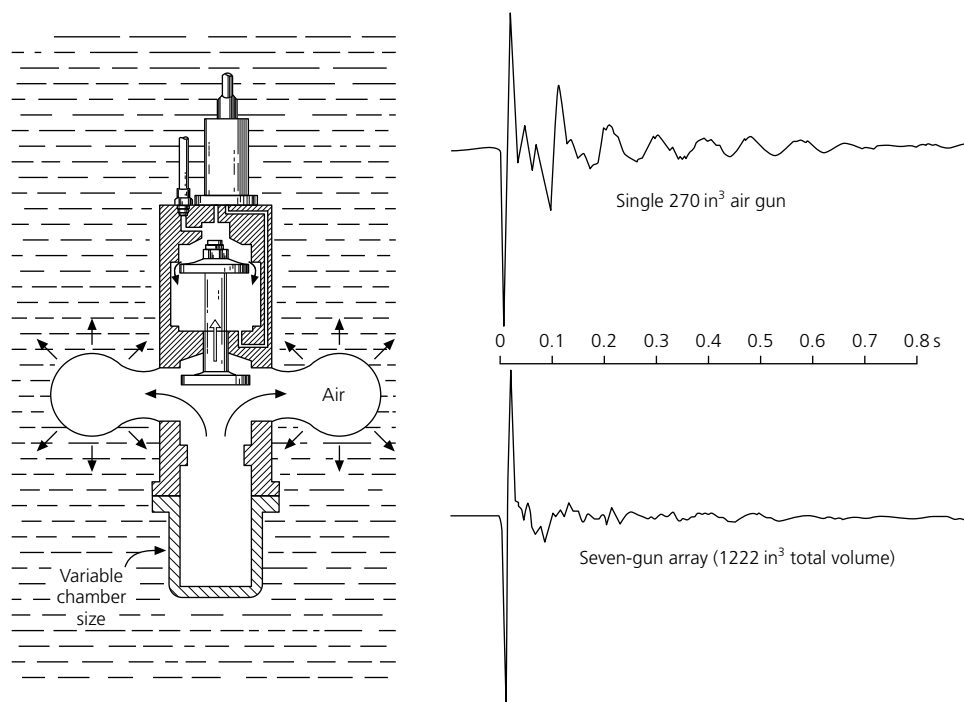


Fig. 3.3-25 *Left*: Schematic of an air gun, a common marine seismic source. (Fig. 3.18 in Kearey and Brooks, 1984, redrawn with permission of Bolt Associates and Soderia Ltd.) *Right*: Source wavelets (pressure versus time) for a single air gun and an array of air guns. The array reduces the bubble pulse and makes the wavelet more impulsive, though it still contains additional unwanted complexity. (Fig. 3.19 in Kearey and Brooks, 1984. Redrawn with permission of Bolt Associates.)

3.3.6 Deconvolution

Another useful technique, *deconvolution*, “sharpens” the reflections from interfaces. Ideally, each reflection would be a sharp pulse approximating a delta function, so the arrival time of the reflection and the depth of the reflector would be deter-

mined precisely. The sharpness of the reflected pulse determines vertical resolution: how close in travel time, and thus depth, two interfaces can be and still give distinct reflected arrivals.

Seismic sources do not generate delta function signals. Figure 3.3-25 shows the signal produced by an air gun, a common source used in marine surveys. The damped oscillation results

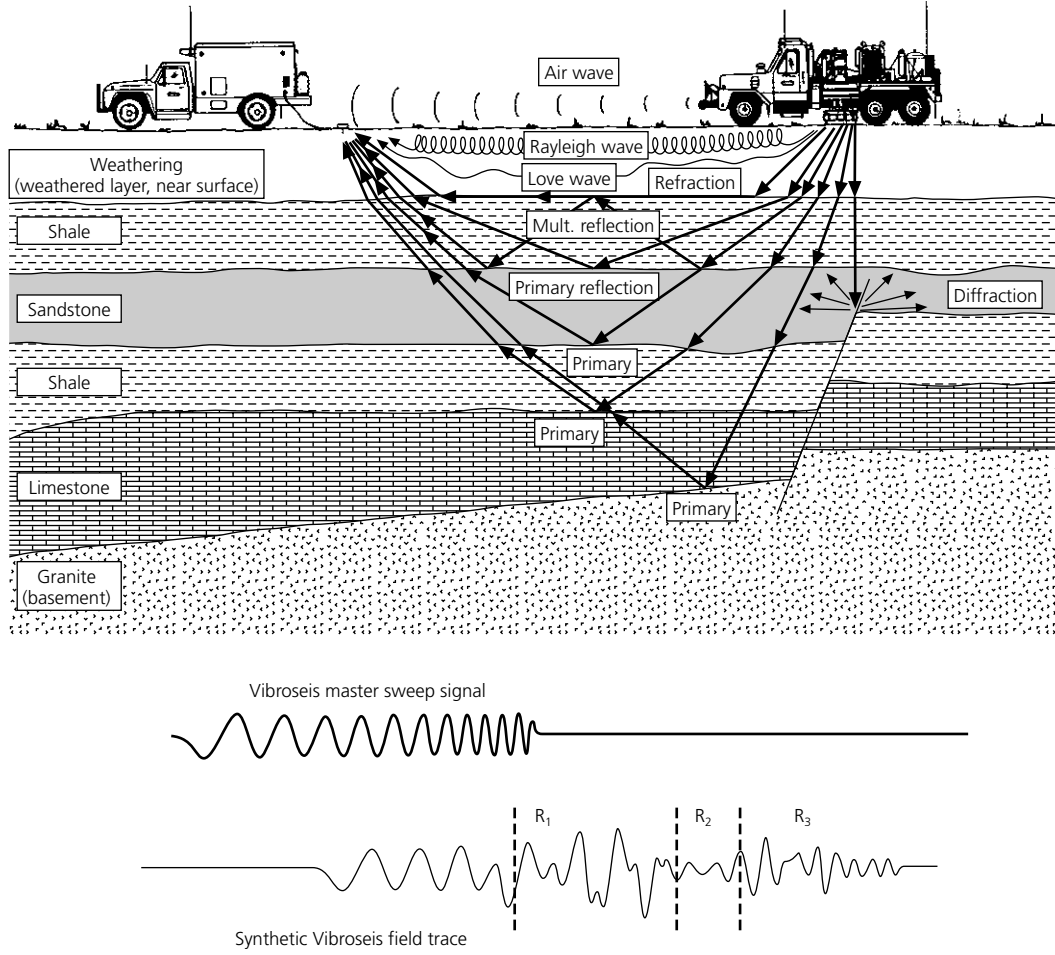


Fig. 3.3-26 Schematic geometry of a Vibroseis survey (top) and sweep signal (center). The field records (bottom) contain interfering reflections off various interfaces, and so require processing to identify individual reflections. (With permission of Conoco.)

from expansion and contraction of the air bubble that the gun injects into the water. The signal can be sharpened using multiple air guns offset in time, which interfere to give a sharper pulse. Figure 3.3-26 shows the “sweep” signal generated by a Vibroseis⁶ unit, a truck-mounted seismic source used in land surveys. The signal extends for a period of time T (typically 7–35 s) over which the frequency varies through a range f_1 – f_2 , generally within 10–60 Hz. Such signals, also called “chirps,” can be written

$$w(t) = \cos 2\pi \left(f_1 t + \frac{(f_2 - f_1)}{2T} t^2 \right). \quad (57)$$

Because the duration of the sweep is often longer than the difference in travel time between interfaces, the resulting seismogram is a complicated combination of sweep signals with different amplitudes and time delays reflected from different interfaces.

Thus reflection data, like any other seismograms, include the effects of both the source and the structure. Separating these effects is a basic theme in seismology, because we are usually interested in either the source (as for earthquakes) or the structure, as in this application. To separate source and structure, we describe a seismogram, $s(t)$, as resulting from the source pulse, known in reflection applications as a wavelet, $w(t)$, and a time series that describes the effects of the structure, in this case a reflector series, $r(t)$.

To find the reflector series, we recall from Section 2.6.7 that a wave with initial unit amplitude acquires an amplitude equal to the product of the reflection and transmission coefficients along its path. Thus, for a set of layers with velocity v_j and thickness h_j , the amplitude of the primary reflection from the bottom of the i^{th} layer is the product of the reflection coefficient at the base of the layer times all the transmission coefficients for both the up and down parts of the path,

$$R_{i+1} \prod_{j=0}^{i-1} T_{j+1} T_{j+1}, \quad (58)$$

⁶ Vibroseis is a trademark of the Continental Oil Company. The first such continuously operating variable frequency seismic source was invented by Selwyn Sacks in his Ph.D. thesis in 1961.

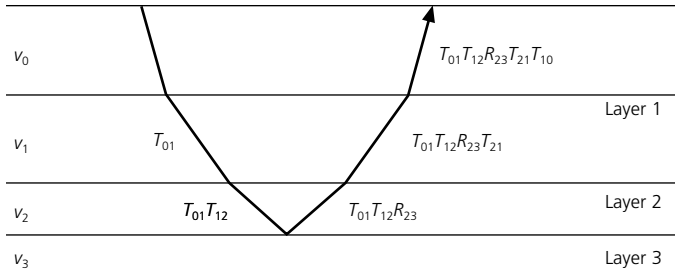


Fig. 3.3-27 Schematic of a ray path through several interfaces, showing how the amplitude depends on the product of the reflection and transmission coefficients along the path.

where Π denotes the product of the indicated terms. For example, the reflection off the base of the second layer has amplitude $R_{23}T_{01}T_{10}T_{12}T_{21} = T_{01}T_{12}R_{23}T_{21}T_{10}$, where the second form shows the order of interactions along the path (Fig. 3.3-27). In dealing with reflection data, the vertical incidence reflection and transmission coefficients are generally suitable approximations. Hence, the reflection and transmission coefficients are given by the densities and velocities at each interface

$$R_{i+1} = \frac{\rho_i v_i - \rho_{i+1} v_{i+1}}{\rho_i v_i + \rho_{i+1} v_{i+1}}, \quad T_{i+1} = \frac{2\rho_i v_i}{\rho_i v_i + \rho_{i+1} v_{i+1}}, \quad (59)$$

and the reflection arrives at a two-way travel time of

$$t_i = 2 \sum_{j=0}^i \frac{h_j}{v_j}, \quad (60)$$

which is the sum of the vertical travel times in each of the layers. Thus the reflector series for primary reflections off a set of N layers is a sum of impulses, each corresponding to the reflection from the bottom of the i^{th} layer,

$$r(t) = \sum_{i=0}^N \delta(t - t_i) R_{i+1} \prod_{j=0}^{i-1} T_{j+1} T_{j+1}. \quad (61)$$

$\delta(t - t_i)$ is the delta function, a spike in time that is zero at all times except t_i , when it equals 1. The reflector series is thus a set of spikes with the appropriate amplitude and arrival time, each corresponding to a specific reflection.

We will see in Chapter 6 that the resulting seismogram is given by an operation known as the *convolution* of $w(t)$ and $r(t)$, which is written

$$s(t) = w(t) * r(t) \equiv \int_{-\infty}^{\infty} w(t - \tau) r(\tau) d\tau. \quad (62)$$

This equation defines convolution in the time domain. Convolution can also be described in the frequency domain,

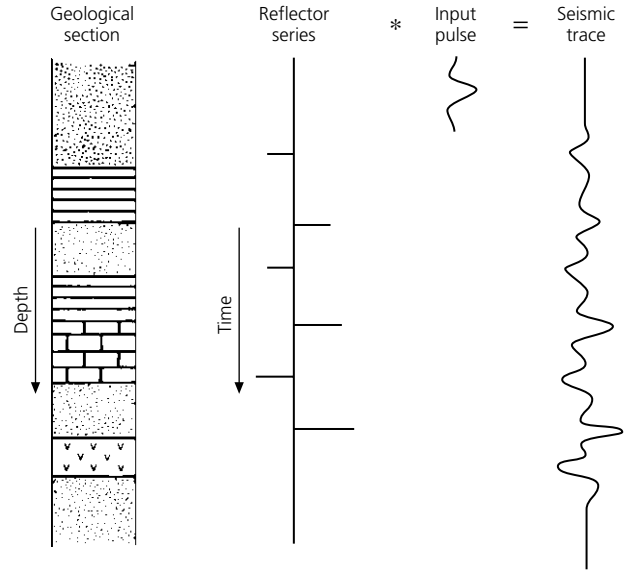


Fig. 3.3-28 A reflection seismogram can be viewed as the convolution of a source wavelet with a reflector series representing the structure. The reflector series has impulses at times corresponding to the arrival times of reflections with amplitudes given by the reflection coefficients. Deconvolution attempts to “spike” the wavelets in the data, revealing the reflector series. (After Kearey and Brooks, 1984.)

because the Fourier transform of a convolution equals the product of the Fourier transforms,

$$S(\omega) = W(\omega)R(\omega). \quad (63)$$

As shown schematically in Fig. 3.3-28, the convolution yields a trace in which the source wavelet appears at times corresponding to the spikes in the reflector series, with the appropriate amplitudes. If the time between the spikes corresponding to individual reflectors is shorter than the duration of the wavelet, interference can give a complicated signal.

These expressions show why it would be desirable to have a delta function source wavelet, because the Fourier transform of a delta function is simply 1. Thus, if $w(t) = \delta(t)$, the seismogram would equal the reflector series. Although a physical source wavelet is not a delta function, the seismograms can be manipulated mathematically to simulate such a wavelet. This can be done by creating an *inverse filter*⁷ $w^{-1}(t)$, that, when convolved with the wavelet, yields a delta function

$$w^{-1}(t) * w(t) = \delta(t). \quad (64)$$

Applying this filter, which “spikes” the wavelet, leaves only the reflector series

$$w^{-1}(t) * s(t) = w^{-1}(t) * w(t) * r(t) = r(t). \quad (65)$$

⁷ The notation $w^{-1}(t)$ does not mean $1/w(t)$.

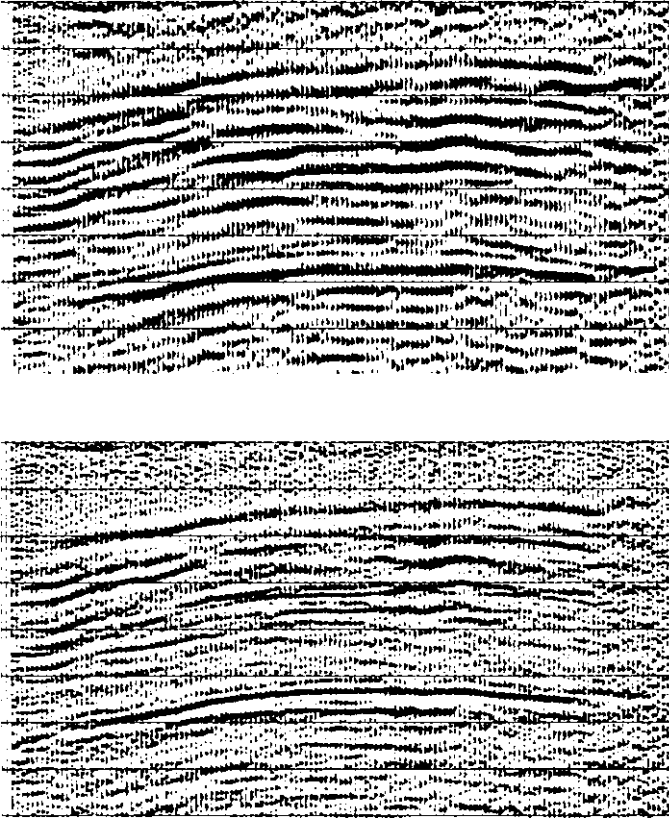


Fig. 3.3-29 *Top*: Seismic section before deconvolution. *Bottom*: Seismic section after deconvolution, showing sharper arrivals for the major reflections. (Yilmaz, 1987. Reproduced by permission of the Society of Exploration Geophysicists.)

Because this operation is the inverse of convolution, it is called *deconvolution*.

To create the inverse filter, note that the Fourier transform of the convolution (Eqn 64) yields

$$W^{-1}(\omega)W(\omega) = 1, \quad (66)$$

so the transform of the inverse filter is just $1/W(\omega)$. Hence deconvolution can be done by dividing the Fourier transforms

$$S(\omega)/W(\omega) = R(\omega). \quad (67)$$

This works well except at frequencies where the source wavelet's spectrum is small. Deconvolution makes the arrivals from reflectors stand out more distinctly (Fig. 3.3-29) and easier to interpret.

An alternative, but similar, approach is used with Vibroseis data for which the wavelet is very long. The goal is to identify times in the trace when the sweep signal arrives. Similarities between two time series $f(t)$ and $g(t)$ are shown by their *cross-correlation*, an operation (Section 6.3.4) defined by

$$c(L) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T}^T f(t+L)g(t)dt. \quad (68)$$

The cross-correlation is largest as a function of L , the lag time, when the series are most similar. For finite time series, the integration is over the times when f and g are nonzero. A special case is the *auto-correlation*, the cross-correlation of a function with itself

$$a(L) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T}^T f(t+L)f(t)dt, \quad (69)$$

which is always maximum at zero lag. The auto-correlation of a Vibroseis sweep, called a Klauder wavelet, is sharply peaked at zero lag (Fig. 3.3-30). Thus cross-correlating a sweep with the recorded trace is similar to using a spiking filter, because it produces sharp spikes when reflections arrive (Fig. 3.3-31). This similarity is not surprising, because cross-correlation and convolution are similar operations (compare Eqns 62 and 68).

Reflections can also be enhanced by filtering in the frequency domain to enhance certain frequency ranges and reject others. The frequency response of geophones varies, but the records may contain frequencies as low as a few Hz and in excess of 100 Hz. As a result, the signal-to-noise ratio can vary significantly as a function of frequency, so filtering often improves reflection quality. The appropriate frequencies may change with time in the record. For example, the later-arriving reflections have longer periods because high-frequency energy is lost to attenuation, the process by which seismic energy is converted to heat (Section 3.7).

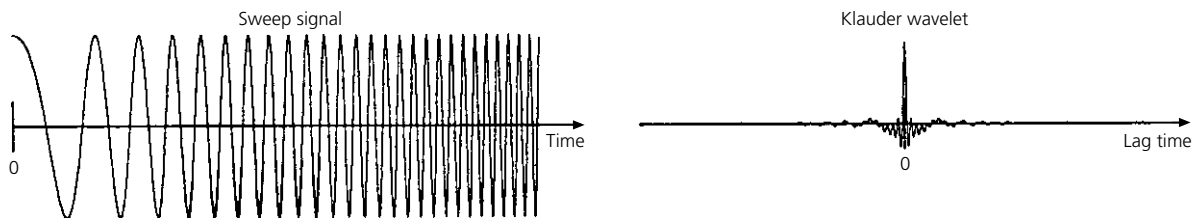


Fig. 3.3-30 The auto-correlation of a Vibroseis sweep signal is an impulsive Klauder wavelet.

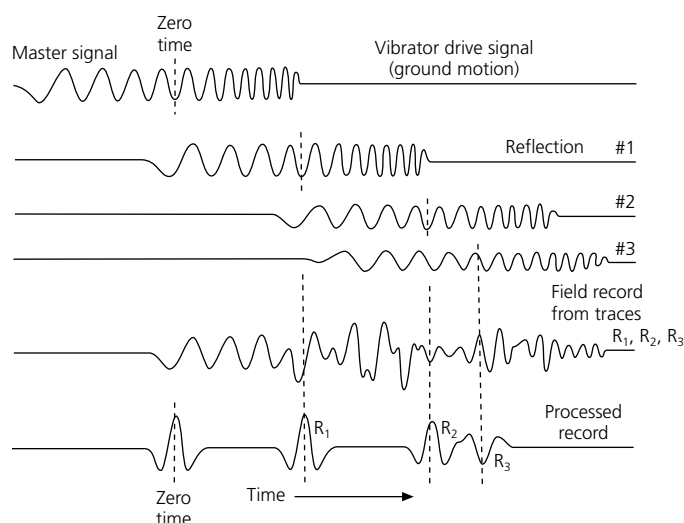


Fig. 3.3-31 A Vibroseis record is a sum of sweep signals reflected from various interfaces. Cross-correlation with a sweep signal produces Klauder wavelets at the reflection times. (Conoco.)

3.3.7 Migration

Given the “cleanest” possible seismic section, how good an image of the subsurface is it? Ideally, the section produced by CMP stacking is a zero-offset section, because the traces have been converted to what would be recorded for a coincident source and receiver. The ray path down to a reflector and back up must be the same, so Snell’s law requires that this path be normally incident on the reflector. If the structure were composed of horizontal interfaces, the reflection paths would be vertical, and the time section could be converted to a depth section by using the velocities to scale the time axis (Fig. 3.3-32, *left*). In this case, a reflection’s arrival time indicates depth to a reflector directly below the source and receiver.

For more complex structures, things get trickier. If interfaces are not horizontal, although the ray paths are the same up and down and intersect the interface at right angles, the path need not be vertical (Fig. 3.3-32, *center*). Moreover, there are several paths from a single source–receiver pair to a reflector. The relation between the zero-offset time section and the structure is thus more complicated.

To deal with these questions, we consider the wave field $u(x, z, t)$, the displacement as a function of position and time during a seismic experiment. The traces are the data at the surface, $u(x, z = 0, t)$. The question of what the traces show about the subsurface can be addressed via a theoretical *exploding reflectors* experiment, in which seismic sources on the reflectors explode at time zero (Fig. 3.3-32, *right*). Waves propagate upward from the reflectors and are recorded at the surface. The reflectors do not interact further with the waves, so multiple reflections are not generated. The sources have strength proportional to the reflection coefficients, so the amplitudes at the surface are correct. Finally, to correct for the fact that the actual reflections went both up and down, times on the recorded traces are divided by two. The recorded data can thus be thought of as resulting from the explosion of the reflectors.

The recorded data are directly related to the structure at depth. At $t = 0$, the instant the sources explode, the wave field at depth, $u(x, z, 0)$, is exactly the geometry of the reflectors, and thus the desired image of the subsurface. These waves propagate upward to the surface $z = 0$, and are recorded as the seismic section $u(x, 0, t)$. Hence the reflectors can be found from the section by removing the effects of propagation, using an operation called *migration*.

We first consider a constant-velocity medium in which a point source at (x_0, z_0) explodes at $t = 0$. The resulting displacement is a circular wave front (Section 2.4.3) that expands with time at a rate equal to the velocity (Fig. 3.3-33) and is described using a delta function

$$u(x, z, t) = \delta((x - x_0)^2 + (z - z_0)^2 - (vt)^2), \quad (70)$$

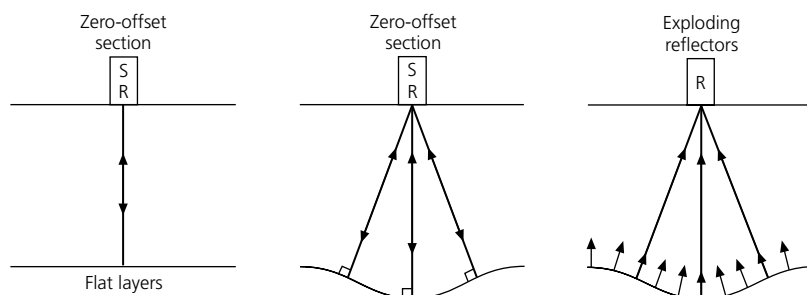


Fig. 3.3-32 Three idealized seismic reflection experiments. *Left*: A zero-offset seismic section for a flat-layered medium. The only reflection points are directly below the source and the receiver. *Center*: A zero-offset seismic section for a medium with a nonhorizontal interface. Although the upgoing and downgoing ray paths are the same, the reflection points need not be directly below the source and the receiver. For a given reflector, several ray paths can produce arrivals at a single receiver. *Right*: A conceptual model in which reflectors explode, giving a wave field with the geometry of the reflectors that propagates to the surface, producing the observed seismic section. Migration seeks to reverse this process and find the initial wave field from the seismic section. (After Claerbout, 1985.)

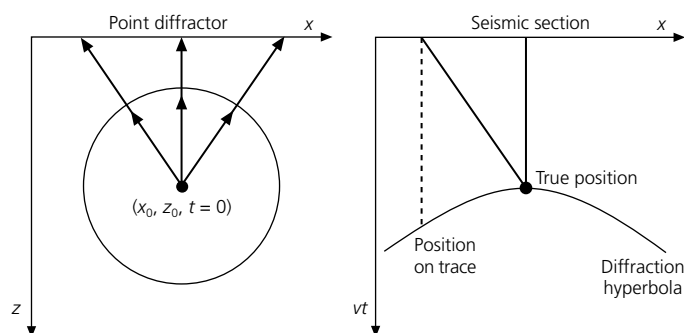


Fig. 3.3-33 Effect of a point source or diffractor. *Left*: The point diffractor acts as a source of spherical (circular in two dimensions) wave fronts. *Right*: The resulting seismic section with time scaled by velocity so that the vertical axis has the dimensions of distance. The diffraction appears as a hyperbola. The point at the apex corresponds to the true position of the diffractor. The other points are due to later arrivals from the diffractor at nonvertical incidence, so their positions on the section do not indicate a reflector directly below the receiver.

whose downgoing half we ignore. The resulting seismic section is the wave field at the surface, $z = 0$,

$$u(x, 0, t) = \delta((x - x_0)^2 + (z_0)^2 - (vt)^2). \quad (71)$$

This is a hyperbola with apex at $(x = x_0, t = z_0/v)$, showing that the wave front arrives first directly above the source, and arrives later at points farther away. Thus the arrival seen on the seismic section is not equivalent to geologic structure with depth. A way to visualize the relation between the source position and the seismic section is to plot the time axis in units of vt , giving a time scale equal to the propagation distance. Thus an arrival time equals the distance along the true path from the source to the receiver. As illustrated, the depth of the source is shown correctly on the section only by the arrival time at a receiver directly above the source. For all other points on the surface, the arrival appears at a time corresponding to the travel time to that point, along a path that was not vertical. Hence, except above the source, the arrival on the section does not correspond to a source directly below the receiver, and the arrival time does not give the source depth directly.

The hyperbolic arrival on a seismic section due to a point source at depth is called a *diffraction hyperbola*. It lets us understand how complicated structures appear on seismic sections, because by Huygens' principle (Section 2.5.10) the reflection from an interface can be found by treating the interface as a set of point sources. The resulting reflection is found by summing the wave fronts from these Huygens' sources, which are also called point diffractors, or point scatterers. Because each source produces a diffraction hyperbola on the seismic section, the section resulting from a set of point diffractors is the sum of their diffraction hyperbolas. In considering this sum, we use the results of a more sophisticated analysis showing that the diffraction hyperbola's amplitude is largest at the apex and decays as the cosine of the angle off to the sides. Be-

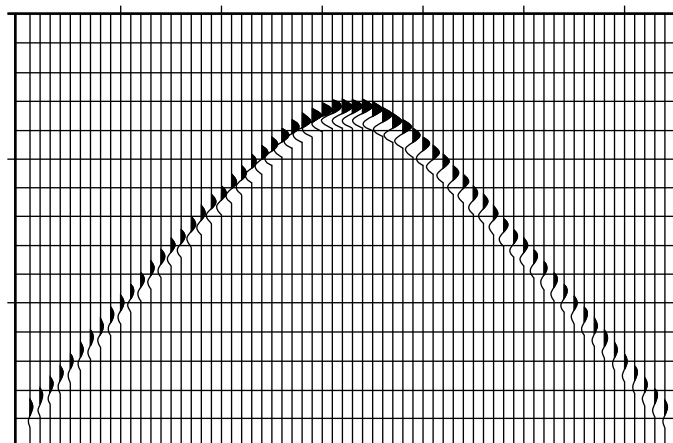


Fig. 3.3-34 Diffraction hyperbola with true amplitudes. (After Claerbout, 1985.)

cause Eqn 71 describes only the travel times, it does not include this term, whose effect is visible in Fig. 3.3-34.

To illustrate this approach, consider an interface dipping at an angle β . This should give the same reflections as a line of closely spaced point diffractors, so the seismic section will be a sum of the resulting hyperbolas. As shown in Fig. 3.3-35, the hyperbolas interfere constructively, causing an apparent interface. Interestingly, this apparent interface does not pass through the apex of each hyperbola, so it is displaced from the real interface and appears to have a shallower dip angle, α . Because the scaled travel time to the true interface equals the scaled arrival time on the trace, the real and apparent dips are related by $\sin \beta = \tan \alpha$.

Seismic sections from simple structures can appear quite different from the actual structure. For example, consideration of ray paths for a single reflector with a synclinal structure shows that several arrivals from different points on the reflector appear on a single zero-offset trace, each with a different travel time. As a result, an apparent anticlinal, or "bowtie," structure appears (Fig. 3.3-36). Another common effect is that the edges of sharp interfaces can give rise to long diffraction "tails" (Fig. 3.3-37). This effect is analogous to diffraction at the edges of a slit (Fig. 2.5-18).

The goal of migration is to undo the effects of diffraction and hence convert the data to realistic images of the subsurface. Migration can thus be thought of as an inverse scattering or inverse diffraction problem. Because this requires removing propagation effects, migration methods are derived using forward models of the propagation process. The idea that the section is the sum of diffractions suggests one approach known as *diffraction sum migration*, or *Kirchoff migration*. Because point diffractors cause hyperbolas on the seismic section, the amplitude of each point on the migrated section is found by summing the unmigrated section with appropriate scaling along hyperbolic trajectories (Fig. 3.3-38). This operation should collapse all the signal in diffraction hyperbolas to points at

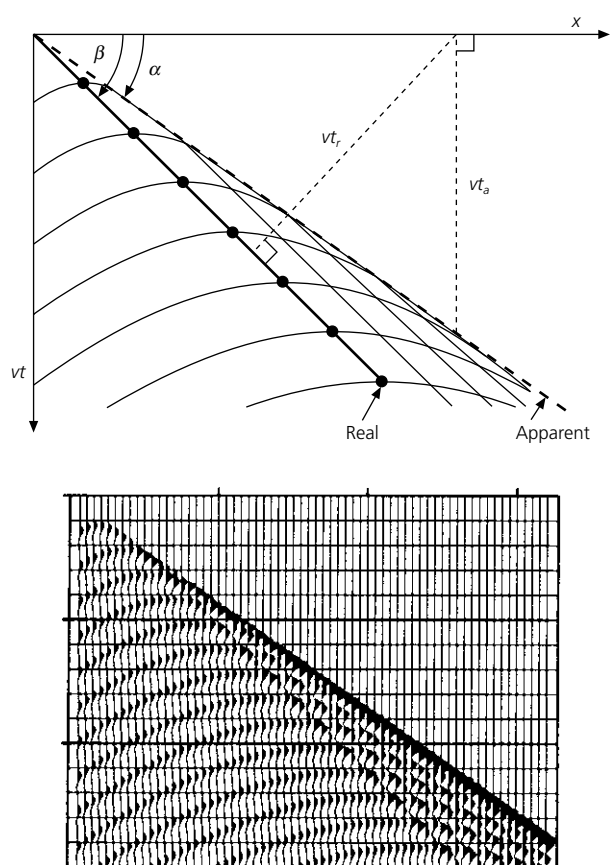


Fig. 3.3-35 A dipping layer can be modeled as a line of point diffractors. On a time section, interference between the diffraction hyperbolas produces an apparent reflector, as shown schematically (*top*) and with true amplitudes (*bottom*). Because the scaled travel time to the true interface (vt_r) equals the scaled arrival time on the trace (vt_a), the apparent dip α is shallower than the true dip β . (After Claerbout, 1985.)

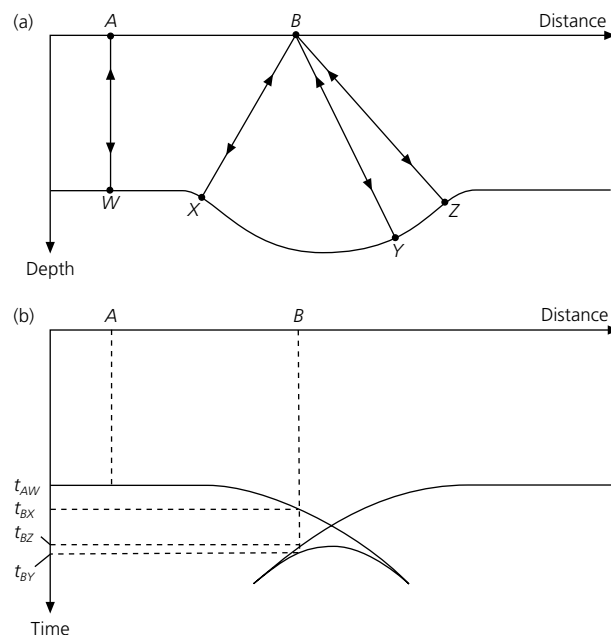


Fig. 3.3-36 *Top*: Illustration of several ray paths for reflections at zero offset from a reflector with a synclinal structure. *Bottom*: Because these arrivals, each from a different point on the reflector, have different travel times, the time section shows an apparent anticlinal, or “bowtie,” structure. (Kearey and Brooks, 1984.)

their apices, and thereby reconstruct the reflectors as a set of point diffractors. Thus diffraction artifacts like those in Figs 3.3-36 and 37 should be removed, and apparent interfaces with shallow dips should be converted to interfaces with the steeper true dips. Figure 3.3-39 shows the improvement to a seismic section from Kirchhoff migration. The resulting migrated time section can be converted to a depth section using a velocity–depth function.

The appearance of a migrated section depends on the assumed velocity. Using a too-slow velocity reduces the length

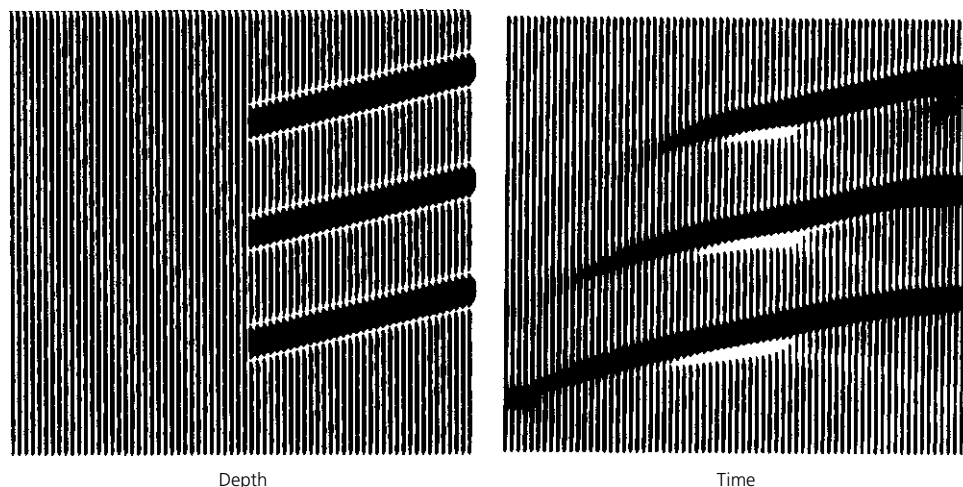


Fig. 3.3-37 The ends of truncated interfaces (*left*) act as diffractors, so a time section (*right*) shows spurious down-dip extensions of the interfaces. (Claerbout, 1976.) (<http://sepwww.stanford.edu/sep/prof/>)

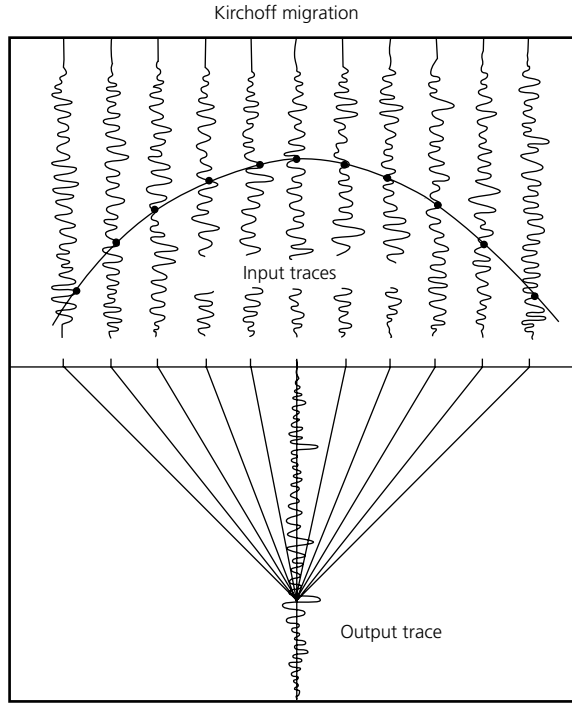


Fig. 3.3-38 Diffraction sum migration reverses the effects of diffraction by summing the time section along hyperbolic trajectories, thus collapsing hyperbolas to their apexes. (Schneider, 1971. Reproduced by permission of the Society of Exploration Geophysicists.)

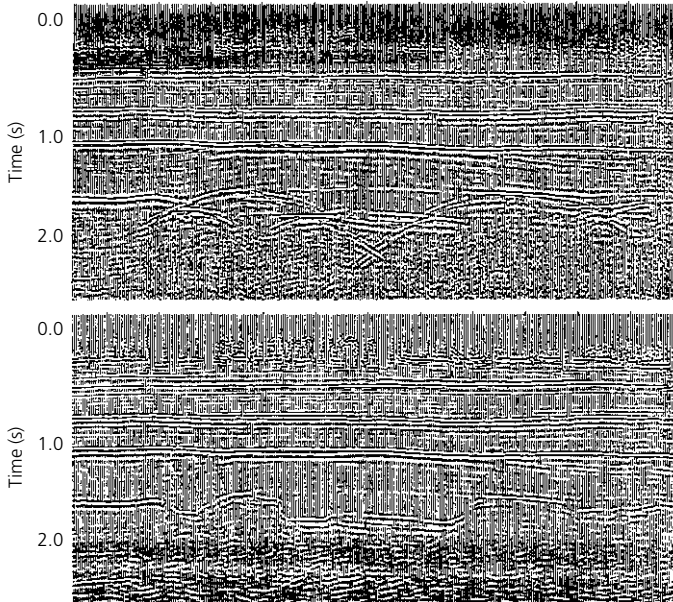


Fig. 3.3-39 Time section before (top) and after (bottom) migration. Elimination of the diffractions produces a better image of structures at depth, such as those at about 1.8 s, where bowties and diffraction “tails” have been suppressed. (Prakla-Seismos)

of hyperbolas’ “tails,” but does not fully collapse them, and so is termed undermigration. Similarly, a too-high velocity overmigrates the data, converting upward-pointing hyperbolas into downward-pointing ones. As a result, correct imaging of dipping structures depends on an accurate velocity model.

Other migration methods, called wave equation migration, use a double Fourier transform to map a wave field, $u(x, z, t)$, from the horizontal distance and time (x, t) domain to the horizontal wavenumber and angular frequency (k_x, ω) domain. The transform is

$$U(k_x, z, \omega) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} u(x, z, t) \exp [i(-\omega t + k_x x)] dx dt, \quad (72)$$

with inverse transform

$$u(x, z, t) = \frac{1}{4\pi^2} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} U(k_x, z, \omega) \exp [i(\omega t - k_x x)] dk_x d\omega. \quad (73)$$

If we consider only P waves, the wave field $u(x, z, t)$ satisfies the wave equation in two dimensions:

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial z^2} = \frac{1}{v^2} \frac{\partial^2 u}{\partial t^2}. \quad (74)$$

The corresponding condition on the transform $U(k_x, z, \omega)$ is found by substituting the inverse transform for u , taking the derivatives, and canceling, yielding

$$\frac{\partial^2 U}{\partial z^2} = \left(k_x^2 - \frac{\omega^2}{v^2} \right) U. \quad (75)$$

Because the components of the wavenumber vector are related by

$$|\mathbf{k}|^2 = k_x^2 + k_z^2 = \omega^2/v^2, \quad (76)$$

the transform satisfies

$$\frac{\partial^2 U}{\partial z^2} = -k_z^2 U. \quad (77)$$

If the velocity is constant with depth, k_z is independent of z , so integrating Eqn 77 yields

$$U(k_x, z, \omega) = U(k_x, 0, \omega) \exp [\pm i k_z z]. \quad (78)$$

This equation relates the wave field at the surface and at any depth. The operation of converting one to the other is called *downward* or *upward continuation* of the wave field. The sign of the exponential distinguishes upcoming from downgoing

waves. Because z increases downward, upcoming waves occur when k_z and ω have the same sign. To ensure this, we define k_z as a function of ω ,

$$k_z(\omega, k_x) = \text{sgn}(\omega) \sqrt{\omega^2/v^2 - k_x^2}, \quad (79)$$

where the function $\text{sgn}(\omega)$ is 1 when ω is positive, and is -1 when ω is negative. Using this definition, the inverse transform Eqn 73 becomes

$$u(x, z, t) = \frac{1}{4\pi^2} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} U(k_x, 0, \omega) \exp [i(\omega t - k_x x + k_z(\omega, k_x)z)] dk_x d\omega. \quad (80)$$

This integral relates the Fourier transform of the seismic section recorded on the surface, $z = 0$, $U(k_x, 0, \omega)$, and the upcoming wave field at depth at earlier times. By the exploding reflector model, the image of the subsurface is the wave field at $t = 0$, when the reflectors have just exploded. Thus the image can be found by setting $t = 0$:

$$u(x, z, 0) = \frac{1}{4\pi^2} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} U(k_x, 0, \omega) \exp [i(-k_x x + k_z(\omega, k_x)z)] dk_x d\omega. \quad (81)$$

Although this integral migrates the transform of the seismic section into the desired image, the integral over ω and k_x has to be done separately to find the image at every depth z . A way to get around this is to replace the ω integration with one over k_z , by expressing ω as a function of k_x and k_z ,

$$\omega(k_x, k_z) = \text{sgn}(k_z) v \sqrt{k_x^2 + k_z^2}, \quad (82)$$

and changing variables using

$$\frac{d\omega}{dk_z} = \frac{k_z v}{\sqrt{k_x^2 + k_z^2}}. \quad (83)$$

This change converts Eqn 81 into an inverse Fourier transform from the wavenumber (k_x, k_z) domain to the space (x, z) domain

$$u(x, z, 0) = \frac{1}{4\pi^2} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} U(k_x, 0, \omega(k_x, k_z)) \exp [i(-k_x x + k_z z)] \frac{k_z v}{\sqrt{k_x^2 + k_z^2}} dk_x dk_z, \quad (84)$$

so inverting the double transform once gives the image for all x and z .

The application of migration methods to data involves various complexities. The time axis in the section can be scaled

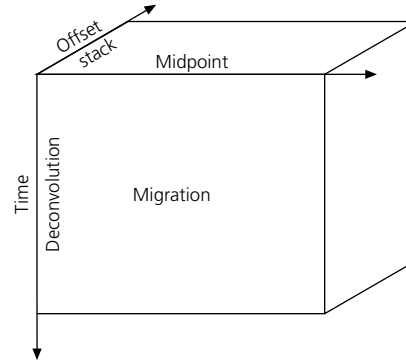


Fig. 3.3-40 Schematic illustration of the relation between processing operations for reflection data. Deconvolution applied along the time axis increases temporal resolution. CMP stacking along the offset axis collapses the data to the midpoint-time plane (compare to Fig. 3.3-18), yielding a seismic section and enhancing reflections. Migration applied in this plane improves lateral resolution. (After Yilmaz, 1987.)

to account for the variation in velocity with depth. Often in complex geology horizontal variations in velocity are important, so migration can be conducted with numerical methods that can propagate waves through laterally varying media.

3.3.8 Data processing sequence

The various processing operations for seismic reflection data can be combined in different ways. To illustrate this, we summarize a common sequence for some of the possible operations. For simplicity, the discussion is in terms of one horizontal dimension, but the approach applies to two dimensions.

Preprocessing consists of initial steps. Because data from different receivers are recorded simultaneously, they are reorganized (demultiplexed) to produce a trace for each receiver. The traces are then edited to eliminate effects such as noisy traces or recording errors. Static time shifts are applied when needed (Section 3.3.5). The amplitudes are then adjusted using a gain recovery function that corrects for the fact that the later arrivals have lower amplitudes because of reflections, transmissions, geometric spreading, and attenuation (Section 3.7). The data are combined into common source gathers, and can then be viewed as a volume defined by the time, offset, and midpoint axes (Fig. 3.3-40). They can then be filtered using methods (Section 3.3.5) including muting of undesired arrivals, band-pass filtering to enhance or suppress certain frequencies, and velocity or slant stack filtering to suppress certain arrivals. Deconvolution (Section 3.3.6), which improves the time resolution of the data, can be viewed as acting along the time axis in Fig. 3.3-40.

As this point, common midpoint stacking and velocity analysis are conducted for the gathers. These operations (Section 3.3.4) combine data for each midpoint to produce a seismic section that approximates what would be recorded at zero offset. Geometrically, this acts along the offset axis to collapse all

the data to the midpoint–time plane. Migration (Section 3.3.7) in the midpoint–time plane seeks to eliminate artifacts due to diffractions and convert the seismic section to an image of the subsurface. The migrated section can then be converted, using assumptions about velocities, to a depth section. The depth section is then interpreted together with geological data and other types of geophysical data, in some cases from drill holes, to understand the subsurface geology.

This discussion of the processing sequence brings out the point that although it is natural to treat a seismic section as an accurate image of the subsurface, it is actually a display of a seismic wave field showing the energy arriving as a function of two-way travel time. Thus the quantity shown, vertical displacement or pressure, need not correspond to any geological reflector of interest. Large arrivals can result from interference between reflections from small impedance contrasts. Moreover, because a seismic section has been produced by mathematical operations, rather than the physical experiment it simulates, noise in the data and errors in the processing can produce spurious artifacts. For example, the conversion of time to depth is only as accurate as the velocities found by stacking or otherwise, perhaps from measurements in a drill hole.

As we have seen in discussing migration, seismic sections are most likely to deviate from the desired images when the medium has significant lateral variations. For example, a medium with random heterogeneities can yield spurious short layered segments, because the reflected energy depends on the vertical changes in impedance. Thus long-wavelength vertical variations in impedance are suppressed, whereas both short- and long-wavelength horizontal variations are preserved, and so can yield a structure with apparent horizontal layering. This effect can be viewed as a velocity filter (Section 3.3.5) that reduces horizontal resolution for structures with steep dips. Similar effects, which are prone to occur at large offsets, may contribute to the horizontally discontinuous layering observed in deep crustal reflection data (Section 3.2.4). Hence, as we will see in various contexts throughout our discussions (e.g., Section 7.3), studying three-dimensional velocity structure is an interesting and challenging enterprise.

3.4 Seismic waves in a spherical earth

In the previous sections, we developed the theory to use the travel times of seismic waves to study the velocity structure of a medium composed of flat layers. This analysis is useful when the ray paths between the source and the receiver are short enough that the earth's curvature can be neglected. Because this is the case for distances less than a few hundred kilometers, such analysis is used to study structure in the crust and the uppermost mantle. In this section we develop the corresponding theory for a spherical earth, which can be used for greater distances and thus greater depths. Application of these results, discussed in the next section, is our primary tool for studying the structure of the deep earth.

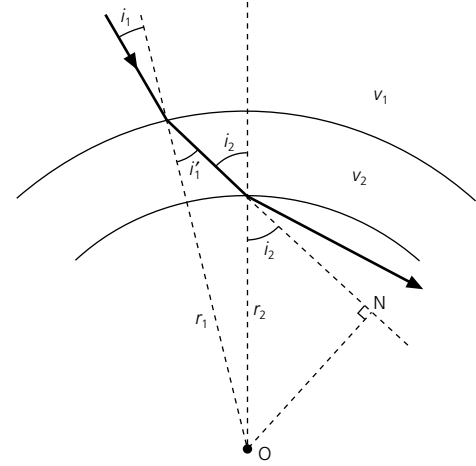


Fig. 3.4-1 Geometry of Snell's law for a spherical earth.

3.4.1 Ray paths and travel times

By analogy to the way we previously represented the earth using uniform flat layers, we now treat it as a series of concentric spherical shells of uniform-velocity material. The ray paths and travel times for the spherical geometry are described by expressions similar to those for flat layers (Section 3.3.1). Consider the portion of a seismic ray's path connecting points at radial distances r_1 and r_2 from the earth's center (Fig. 3.4-1). If v_1 and v_2 are the velocities above and below r_1 , and i_1 , i_1' and i_2 are the angles shown, then by Snell's law

$$\frac{r_1 \sin i_1}{v_1} = \frac{r_1 \sin i_1'}{v_2}. \quad (1)$$

However, $r_1 \sin i_1' = r_2 \sin i_2$ because both equal the length ON , so we rewrite Eqn 1 as

$$\frac{r_1 \sin i_1}{v_1} = \frac{r_2 \sin i_2}{v_2}. \quad (2)$$

Thus we define the ray parameter p for a spherical earth as

$$p = \frac{r \sin i}{v}, \quad (3)$$

where r is the radial distance from the center of the earth, v is the velocity at that point, and i is the incidence angle between the ray path and the radius vector. By reducing the thickness of the shells ever thinner, the velocity becomes a continuous function of radius, $v(r)$. Equation 3 is thus Snell's law for a spherical earth, which describes the ray path. As for the flat earth, the ray parameter is constant along the ray path, and thus identifies a particular ray.

It may seem strange that different forms of the ray parameter and Snell's law occur for a sphere. At any given depth, the flat